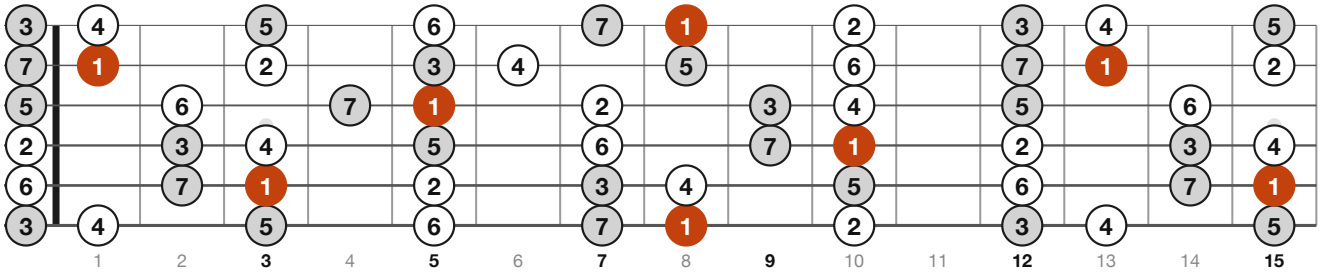
GUITAR CARDS · STUDY GUIDE

Guitar Mastery

The fretboard, harmony and practice in one place: cheat sheets first, then every essential shape on the neck, then how to turn it into music.



Every note spelling, interval label and fret position in this guide is generated by the Guitar Cards theory engine, so the diagrams are correct by construction. Examples are in C (or the most convenient key); every shape is movable — slide it so its root lands on the root you need.

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PART II · THE NECK

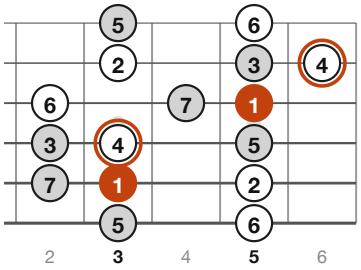
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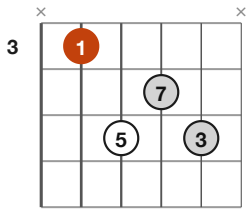
How to read and use this guide

Neck diagrams (scales, arpeggios, maps)



- Strings run **horizontally**: high e on top, low E at the bottom — the neck as you see it when you look down at it while playing.
- Fret numbers under the neck; inlay dots at 3, 5, 7, 9, 12, 15. A thick left edge is the nut; notes left of it are open strings.
- Example: C major, A-shape window. Ring = the characteristic note being discussed (here the 4th, the avoid note over Cmaj7).

Chord boxes (voicings)



- Strings run **vertically**: low E on the **left**, high e on the right; frets go down the page.
- The number at the top left is the fret of the first row (no number = nut / first position).
- × = string not played.
- Example: Cmaj7 drop 2, root position on strings 5-4-3-2.

Dot colours and labels

1 root 3 guide / chord tone 6 other note #4 characteristic note

- Every dot is labelled with its **interval from the root** (1, ♭3, 3, 5, ♭7, 9, ♯11, 13…), never with a fingering. Saying the interval as you play is the point.
- **Grey** in chord boxes = the guide tones (3rd and 7th), the notes that define the chord. **Grey** in scale diagrams = the notes of the scale's own 7th chord (the arpeggio hidden in the scale).
- Roman numerals are always relative to the **major** scale of the tonic: a minor key reads i iiø ♭III iv v ♭VI ♭VII.
- Compound tensions (9, 11, 13) name the same notes as 2, 4, 6 an octave higher; in scale diagrams the engine labels scale degrees (2, 4, 6) unless the scale is normally thought of as tensions (altered: ♭9 ♯9 ♯11 ♭13).

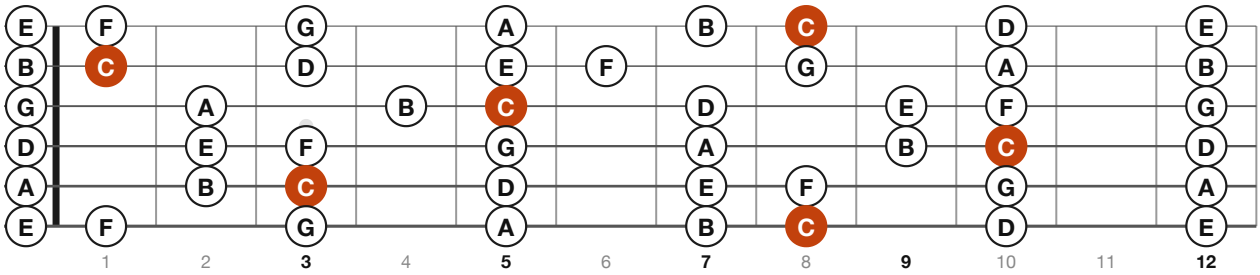
How to use it

- **Part I** is for lookup: keep it open while you practise and when you analyse tunes.
- **Part II** is the hands: work through it one row at a time, in the order given, and transpose every shape to at least three other keys (then all 12) before you call it learned.
- **Part III** turns shapes into music: voice leading, improvising, rhythm and ear. Section 27 is the training plan and the checklist that tells you when each section is mastered.
- Mastery means: you can find it **without looking**, **in any key**, **in time**, and **hear it** before you play it.

1 The fretboard: notes and octaves

Standard tuning E A D G B E (low to high). All diagrams in this guide show the neck as you look down at it: high e string on top, low E at the bottom, nut on the left.

Natural notes, frets 0–12 (C highlighted; sharps/flats sit between; E–F and B–C are one fret apart)



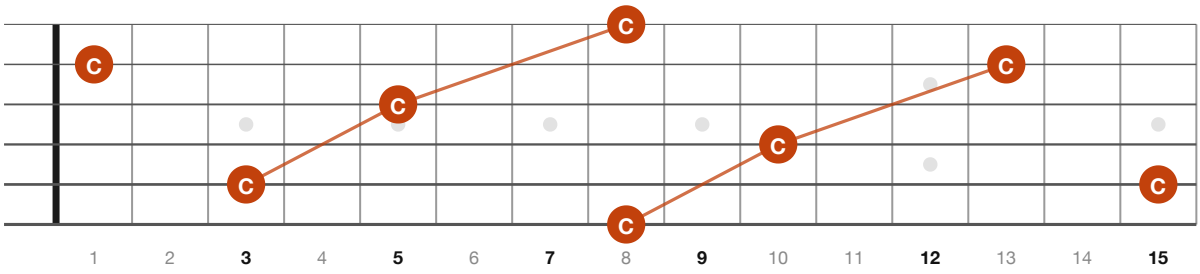
Octave shapes

Root on	Octave	Move
string 6 → string 4	one octave	+2 frets
string 5 → string 3	one octave	+2 frets
string 4 → string 2	one octave	+3 frets (crosses G–B: one fret further)
string 3 → string 1	one octave	+3 frets (crosses G–B: one fret further)
string 6 → string 1	two octaves	same fret (crosses G–B: one fret further)
string 5 → string 2	one octave	–2 frets (crosses G–B: one fret further)
string 6 → string 3	one octave	–3 frets
string 4 → string 1	one octave	–2 frets (crosses G–B: one fret further)

Landmarks

- Frets 5, 7, 12 are your anchors: fret 12 repeats the open strings an octave up.
- Fret 5 on any string = the next (higher) open string, **except** the G string: fret 4 = open B.
- Learn strings 6 and 5 first (every barre-chord and CAGED root lives there), then use octave shapes to find the same note on strings 4–1.
- Same note, next string up: **5 frets lower** (4 frets lower crossing G→B).
- Drill: choose a note, find it on all six strings low→high→low in under 10 seconds, saying the string and fret aloud. One note per day, all 12 in two weeks.

Every C from fret 0 to 15 with the octave shapes joined (two strings up: +2 frets, or +3 when the pair straddles the B string)



2 Intervals: names, sounds and shapes

Interval	Label	Semis	Inverts to	Same string	Next string up*	Ascending reference	Descending reference	Character
m2	**♭2**	1	M7	+1	−4	The Entertainer (opening D–D♯–E)	Joy to the World (first two notes)	tense, rubbing
M2	**2**	2	m7	+2	−3	Frère Jacques	Mary Had a Little Lamb	a step
m3	**♭3**	3	M6	+3	−2	Greensleeves	This Old Man	minor, sad
M3	**3**	4	m6	+4	−1	When the Saints Go Marching In	Swing Low, Sweet Chariot	major, bright
P4	**4**	5	P5	+5	0 (same fret)	Amazing Grace; Auld Lang Syne	O Come, All Ye Faithful	open, hymn-like
TT	**♯4/♭5**	6	TT	+6	+1	Saint-Saëns, Danse Macabre (violin)	same (it is symmetric)	unstable; 3rd + ♭7 of a V7
P5	**5**	7	P4	+7	+2	Twinkle Twinkle Little Star	Minuet in G (Petzold, opening D–G)	strong, hollow
m6	**♭6**	8	M3	+8	+3	The Entertainer (E up to C)	—	bittersweet
M6	**6**	9	m3	+9	+4	My Bonnie Lies Over the Ocean	Nobody Knows the Trouble I've Seen	sweet, open
m7	**♭7**	10	M2	+10	+5 · or 2 strings up, same fret	root → ♭7 of any dominant 7th	—	bluesy, wants to fall
M7	**7**	11	m2	+11	+6 · or 2 strings up, −1	one fret below the octave	—	lush, leading tone
P8	**8**	12	P1	+12	2 strings up, +2 (+3 over B)	same note, higher	Joy to the World (whole first phrase)	the same note

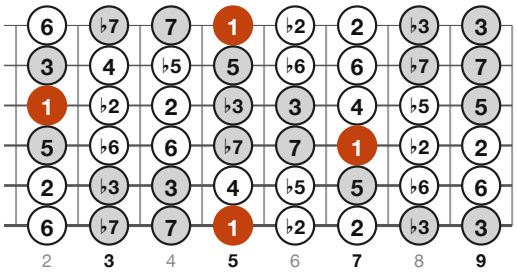
* Moving to the next higher string, not crossing G→B. Crossing from G to B, add one fret. Interval reference tunes are all traditional or public domain; where none is reliable the chord-sound reference is given.

Compound	♭9	9	♯9	11	♯11	♭13	13
= simple + octave	♭2	2	♯2 (sounds as ♭3)	4	♯4 (= ♭5)	♭6 (= ♯5)	6

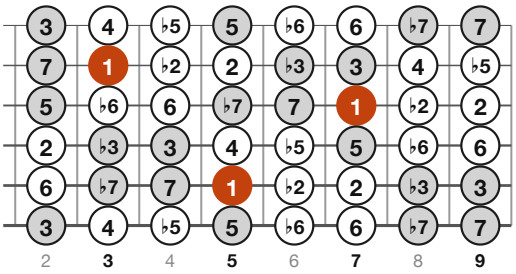
Every interval around a root — learn these four maps and you can find any interval anywhere

Accent = root. Grey = the chord tones (♭3, 3, 5, ♭7, 7). Compare maps 1–2 with 3–4: any shape that crosses the G–B pair shifts up one fret on the B and e strings.

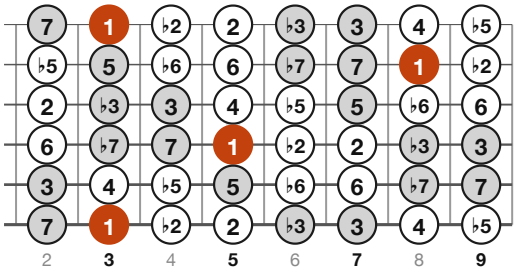
Root on string 6 (A, fret 5)



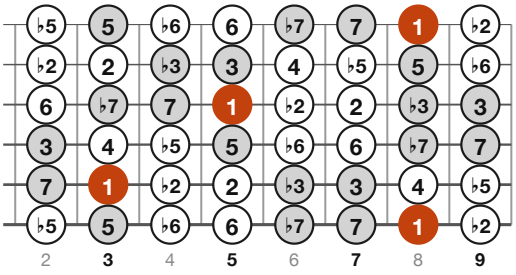
Root on string 5 (D, fret 5)



Root on string 4 (G, fret 5)



Root on string 3 (C, fret 5)



3 Chord formulas

Every chord in one table, spelled on C. To use another root, keep the formula and move the shape.

Triads and power chords

Chord (on C)	Also written	Formula	Notes	Typical job
C	M, maj	**1 3 5**	C E G	I, IV, V of a major key
Cm	m, min, –	**1 ♭3 5**	C E♭ G	ii, iii, vi; minor tonic
C°	°, dim	**1 ♭3 ♭5**	C E♭ G♭	vii° in major, ii° in minor
C+	+, aug	**1 3 ♯5**	C E G♯	passing (I → I+ → vi/IV), V+
Csus2		**1 2 5**	C D G	colour, no 3rd
Csus4		**1 4 5**	C F G	Vsus4 → V; decoration
C5	power chord	**1 5**	C G	rock riffs (neither major nor minor)

Sixths and added-note chords

Chord (on C)	Also written	Formula	Notes	Typical job
C6	M6, add6	**1 3 5 6**	C E G A	major tonic (swing, bossa)
Cm6	–6	**1 ♭3 5 6**	C E♭ G A	minor tonic; iv6 in major
Cadd9		**1 3 5 9**	C E G D	pop colour on I, IV
Cm(add9)		**1 ♭3 5 9**	C E♭ G D	minor colour
C6/9	69	**1 3 5 6 9**	C E G A D	final major tonic
Cm6/9		**1 ♭3 5 6 9**	C E♭ G A D	final minor tonic

Seventh chords

Chord (on C)	Also written	Formula	Notes	Typical job
Cmaj7	Δ7, M7	**1 3 5 7**	C E G B	I, IV
C7	dom7	**1 3 5 ♭7**	C E G B♭	V; every chord of a blues
Cm7	–7, min7	**1 ♭3 5 ♭7**	C E♭ G B♭	ii, iii, vi; Dorian i
Cm7♭5	ø7, m7(♭5)	**1 ♭3 ♭5 ♭7**	C E♭ G♭ B♭	vii in major; ii in minor
C°7	°7, dim7	**1 ♭3 ♭5 𝄫7**	C E♭ G♭ B𝄫	vii°7 in minor; passing chord
Cm(maj7)	m(Δ7), –Δ7	**1 ♭3 5 7**	C E♭ G B	minor tonic (melodic/harmonic minor)
Cmaj7♯5	+maj7, Δ♯5	**1 3 ♯5 7**	C E G♯ B	♭III+ in minor; tonic colour
C7♯5	+7, aug7, 7(♯5)	**1 3 ♯5 ♭7**	C E G♯ B♭	V whose #5 rises to the next 3rd
C7♭5	7(♭5)	**1 3 ♭5 ♭7**	C E G♭ B♭	V; same notes as its tritone sub
C7sus4		**1 4 5 ♭7**	C F G B♭	V delayed; modal vamp

Extended chords (9, 11, 13)

Chord (on C)	Also written	Formula	Notes	Typical job
Cmaj9	Δ9	**1 3 5 7 9**	C E G B D	I, IV
C9		**1 3 5 ♭7 9**	C E G B♭ D	V; funk vamp
Cm9	–9	**1 ♭3 5 ♭7 9**	C E♭ G B♭ D	ii; minor tonic
C11	9sus4 (no 3rd)	**1 5 ♭7 9 11**	C G B♭ D F	sus dominant (= IV/V)
Cm11		**1 ♭3 5 ♭7 9 11**	C E♭ G B♭ D F	Dorian vamp; ii
Cmaj7♯11	Δ♯11, Lydian chord	**1 3 5 7 ♯11**	C E G B F♯	IV; Lydian tonic
C13		**1 3 5 ♭7 9 13**	C E G B♭ D A	V (Mixolydian)
Cm13		**1 ♭3 5 ♭7 9 13**	C E♭ G B♭ D A	Dorian i / ii
Cmaj13	Δ13	**1 3 5 7 9 13**	C E G B D A	I, IV
C9sus4		**1 4 5 ♭7 9**	C F G B♭ D	V sus
C13sus4		**1 4 5 ♭7 9 13**	C F G B♭ D A	V sus

Altered dominants

Chord (on C)	Also written	Formula	Notes	Typical job
C7♭9		**1 3 5 ♭7 ♭9**	C E G B♭ D♭	V → minor i
C7♯9		**1 3 5 ♭7 ♯9**	C E G B♭ D♯	blues/rock I7; V alt
C7♯11	7(♯11), 9♯11	**1 3 5 ♭7 9 ♯11**	C E G B♭ D F♯	II7, ♭VII7, ♭II7 (non-resolving V)
C7♭13		**1 3 5 ♭7 ♭13**	C E G B♭ A♭	V → minor
C7♭9♭13		**1 3 5 ♭7 ♭9 ♭13**	C E G B♭ D♭ A♭	V → minor (harmonic minor)

Chord (on C)	Also written	Formula	Notes	Typical job
C7♯9♭13		**1 3 5 ♭7 ♯9 ♭13**	C E G B♭ D♯ A♭	altered V
C7♯9♯5		**1 3 ♯5 ♭7 ♯9**	C E G♯ B♭ D♯	altered V
C13♭9	13(♭9)	**1 3 ♭7 ♭9 13**	C E B♭ D♭ A	V (half-whole diminished)
C13♯11	13(♯11)	**1 3 ♭7 9 ♯11 13**	C E B♭ D F♯ A	II7, ♭II7 (Lydian dominant)
C7alt	7(♯9♭13), alt	**1 3 ♭7 ♭9 ♯9 ♯11 ♭13**	C E B♭ D♭ D♯ F♯ A♭	V → I or i, maximum tension

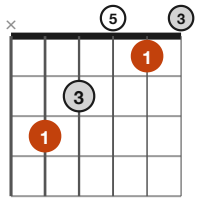
Rules that generate every chord symbol

- **Triad** = 1 3 5 stacked in thirds; the 3rd decides major/minor, the 5th decides diminished/augmented.
- **7th chords** add a 3rd on top: maj7 = major 7th (11 semitones), 7 = minor 7th (10), °7 = diminished 7th (9 = same pitch as 6).
- **A plain "7" always means ♭7.** "maj" in a symbol only ever refers to the 7th.
- **9, 11, 13** imply the 7th (C9 contains B♭); **add9 / 6** chords have no 7th. "sus" replaces the 3rd with 4 (or 2).
- **Tensions** (9, 11, 13 and their alterations) are the scale notes between chord tones, numbered an octave up. Avoid the natural 11 over a major 3rd unless it is a passing note (use #11 instead).
- On guitar the **5th is the first note to drop**, then the root (a bassist or the ear supplies it). The 3rd and 7th define the chord: keep them.

4 Essential chord grips: open and barre

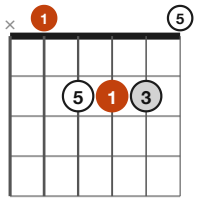
Open-position chords

C



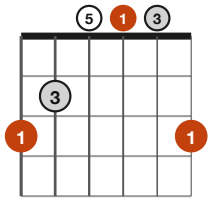
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A



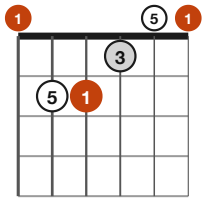
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G



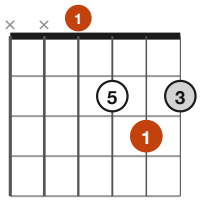
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E



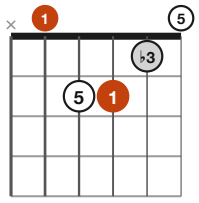
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D



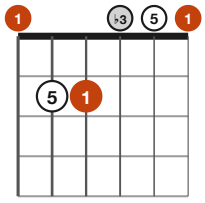
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Am



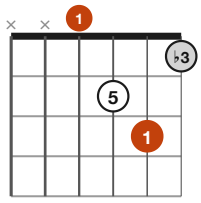
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Em



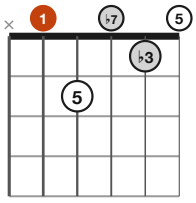
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Dm



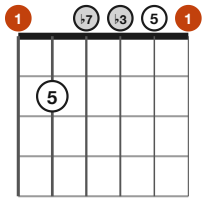
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Am7



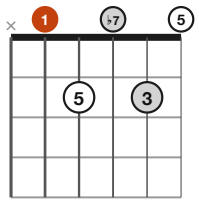
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Em7



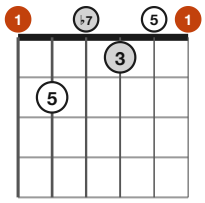
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A7



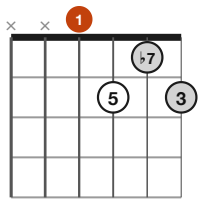
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E7



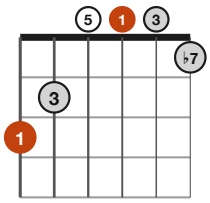
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D7



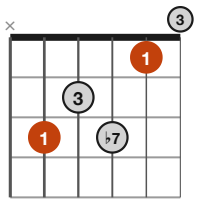
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G7



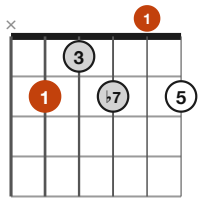
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C7



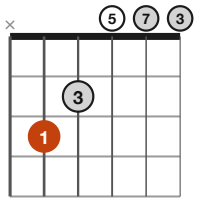
× 3 2 3 1 0

B7



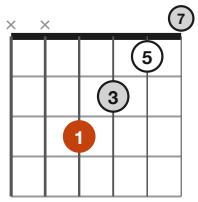
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Cmaj7



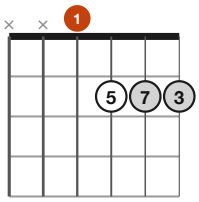
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Fmaj7



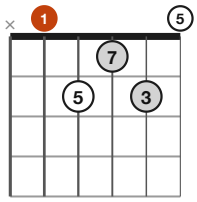
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Dmaj7



× × 0 2 2 2

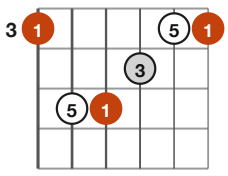
Amaj7



× 0 2 1 2 0

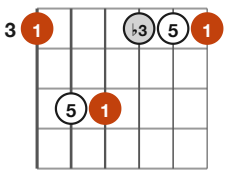
E-shape barre chords (root on string 6) — shown on G, 3rd fret

G



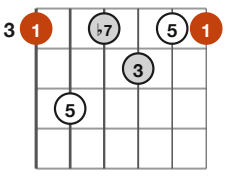
E shape · root fret 3

Gm



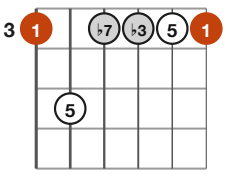
E shape · root fret 3

G7



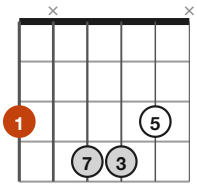
E shape · root fret 3

Gm7



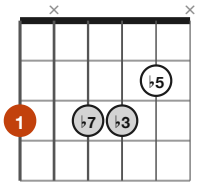
E shape · root fret 3

Gmaj7



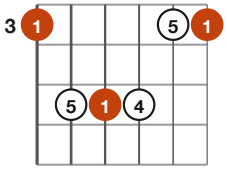
E shape · root fret 3

Gm7♭5



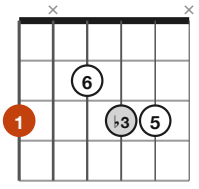
E shape · root fret 3

Gsus4



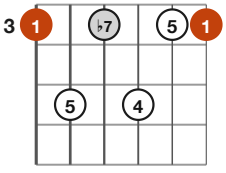
E shape · root fret 3

Gm6



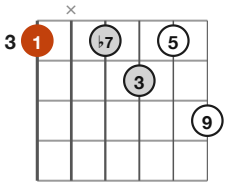
E shape · root fret 3

G7sus4



E shape · root fret 3

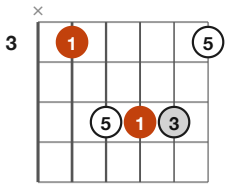
G9



E shape · root fret 3

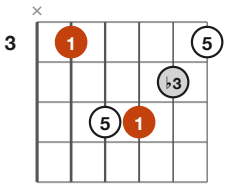
A-shape barre chords (root on string 5) — shown on C, 3rd fret

C



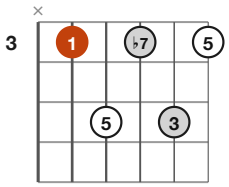
A shape · root fret 3

Cm



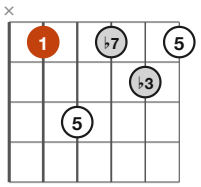
A shape · root fret 3

C7



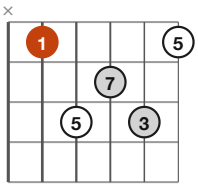
A shape · root fret 3

Cm7



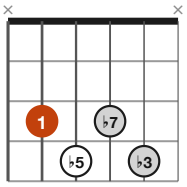
A shape · root fret 3

Cmaj7



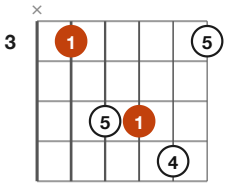
A shape · root fret 3

Cm7♭5



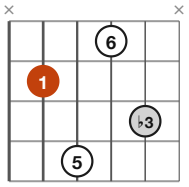
A shape · root fret 3

Csus4



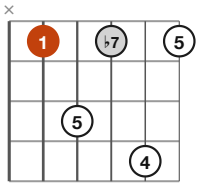
A shape · root fret 3

Cm6



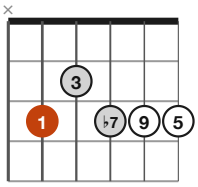
A shape · root fret 3

C7sus4



A shape · root fret 3

C9



A shape · root fret 3

Move a barre shape so the accent dot sits on the root you need: string 6 roots use the E shapes, string 5 roots the A shapes (section 1 shows where every root is). Grey = 3rd and 7th: compare the shapes and notice that changing quality only moves those notes.

5 Harmonic systems: chords and modes

Each parent scale, harmonised in 7th chords. Columns: the chord built on each degree (roman numerals relative to the major scale of the tonic), the mode that starts on that degree, its formula measured against the major scale of its own root, the note that gives it its flavour, and where you would use it.

Major scale system (the seven "church" modes) — on C: C D E F G A B

°	Chord (in C)	Mode (starts on this degree)	Formula vs major	Characteristic note	Typical use
1	**Imaj7** Cmaj7 triad I	**C Ionian (major)**	**1 2 3 4 5 6 7**	7 and natural 4	Imaj7 (4 = avoid note over the 3rd)
2	**ii7** Dm7 triad ii	**D Dorian**	**1 2 ♭3 4 5 6 ♭7**	♮6	ii7; modal/funk minor i7
3	**iii7** Em7 triad iii	**E Phrygian**	**1 ♭2 ♭3 4 5 ♭6 ♭7**	♭2	iii7; Spanish/metal minor; sus♭9
4	**IVmaj7** Fmaj7 triad IV	**F Lydian**	**1 2 3 ♯4 5 6 7**	♯4	IVmaj7; any maj7♯11
5	**V7** G7 triad V	**G Mixolydian**	**1 2 3 4 5 6 ♭7**	♭7	V7 → I; blues I7; I–♭VII–IV rock
6	**vi7** Am7 triad vi	**A Aeolian (natural minor)**	**1 2 ♭3 4 5 ♭6 ♭7**	♭6	vi7; minor-key i; i–♭VI–♭VII
7	**viiø7** Bm7♭5 triad vii°	**B Locrian**	**1 ♭2 ♭3 4 ♭5 ♭6 ♭7**	♭5 (and ♭2)	viiø7 of a major key

Melodic minor system (jazz minor: the ascending form) — on C: C D E♭ F G A B

°	Chord (in C)	Mode (starts on this degree)	Formula vs major	Characteristic note	Typical use
1	**i(maj7)** Cm(maj7) triad i	**C Melodic minor**	**1 2 ♭3 4 5 6 7**	♮7 under a ♭3	im(maj7), im6 tonic
2	**ii7** Dm7 triad ii	**D Dorian ♭2**	**1 ♭2 ♭3 4 5 6 ♭7**	♭2 with ♮6	7sus4♭9 (Phrygian sus)
3	**♭III+maj7** E♭maj7♯5 triad ♭III+	**E♭ Lydian augmented**	**1 2 3 ♯4 ♯5 6 7**	♯5 (and ♯4)	maj7♯5
4	**IV7** F7 triad IV	**F Lydian dominant**	**1 2 3 ♯4 5 6 ♭7**	♯4 with ♭7	7♯11: II7, ♭VII7, ♭II7, IV7 in minor
5	**V7** G7 triad V	**G Mixolydian ♭6**	**1 2 3 4 5 ♭6 ♭7**	♭6 with 3	7♭13 → minor
6	**viø7** Am7♭5 triad vi°	**A Locrian ♮2**	**1 2 ♭3 4 ♭5 ♭6 ♭7**	♮2 with ♭5	iiø7 of a minor ii–V
7	**viiø7** Bm7♭5 triad vii°	**B Altered**	**1 ♭9 ♯9 3 ♯11 ♭13 ♭7**	♭9 ♯9 ♯11 ♭13	V7alt → I or i

Harmonic minor system — on C: C D E♭ F G A♭ B

°	Chord (in C)	Mode (starts on this degree)	Formula vs major	Characteristic note	Typical use
1	**i(maj7)** Cm(maj7) triad i	**C Harmonic minor**	**1 2 ♭3 4 5 ♭6 7**	♮7 with ♭6 (augmented 2nd)	im(maj7) with a leading tone
2	**iiø7** Dm7♭5 triad ii°	**D Locrian ♮6**	**1 ♭2 ♭3 4 ♭5 6 ♭7**	♮6 with ♭5	iiø7 in minor
3	**♭III+maj7** E♭maj7♯5 triad ♭III+	**E♭ Ionian ♯5**	**1 2 3 4 ♯5 6 7**	♯5	♭III+maj7 in minor
4	**iv7** Fm7 triad iv	**F Dorian ♯4**	**1 2 ♭3 ♯4 5 6 ♭7**	♯4 with ♭3	iv7 in minor; Ukrainian/klezmer
5	**V7** G7 triad V	**G Phrygian dominant**	**1 ♭9 3 4 5 ♭13 ♭7**	♭2 with 3	V7♭9♭13 → minor; flamenco
6	**♭VImaj7** A♭maj7 triad ♭VI	**A♭ Lydian ♯2**	**1 ♯2 3 ♯4 5 6 7**	♯2 with ♯4	♭VImaj7 in minor (exotic)
7	**vii°7** B°7 triad vii°	**B Altered ♭♭7 (superlocrian ♭♭7)**	**1 ♭2 ♭3 ♭4 ♭5 ♭6 𝄫7**	𝄫7 (= 6) with ♭4 (= 3)	vii°7 → i

Harmonic major system — on C: C D E F G A♭ B

°	Chord (in C)	Mode (starts on this degree)	Formula vs major	Characteristic note	Typical use
1	**Imaj7** Cmaj7 triad I	**C Harmonic major**	**1 2 3 4 5 ♭6 7**	♭6 with 3	Imaj7 with a ♭6 colour
2	**iiø7** Dm7♭5 triad ii°	**D Dorian ♭5**	**1 2 ♭3 4 ♭5 6 ♭7**	♭5 with ♮6	iiø7 → V7 → I (major)
3	**iii7** Em7 triad iii	**E Phrygian ♭4**	**1 ♭2 ♭3 ♭4 5 ♭6 ♭7**	♭4 (= 3) with ♭2	iii7, rarely
4	**iv(maj7)** Fm(maj7) triad iv	**F Lydian ♭3**	**1 2 ♭3 ♯4 5 6 7**	♭3 with ♯4	iv(maj7) borrowed in major
5	**V7** G7 triad V	**G Mixolydian ♭2**	**1 ♭2 3 4 5 6 ♭7**	♭2 with 3 and ♮6	V7♭9 → major I
6	**♭VI+maj7** A♭maj7♯5 triad ♭VI+	**A♭ Lydian augmented ♯2**	**1 ♯2 3 ♯4 ♯5 6 7**	♯2 ♯4 ♯5	♭VI+maj7 (rare)
7	**vii°7** B°7 triad vii°	**B Locrian ♭♭7**	**1 ♭2 ♭3 4 ♭5 ♭6 𝄫7**	𝄫7 with ♭2	vii°7 → major I

Parallel view: all seven modes on the same root, brightest to darkest

Mode on C	Notes	Formula	Change from the line above	Same notes as
C Lydian	C D E F♯ G A B	1 2 3 ♯4 5 6 7		G major
C Ionian (major)	C D E F G A B	1 2 3 4 5 6 7	♯4 → 4	C major
C Mixolydian	C D E F G A B♭	1 2 3 4 5 6 ♭7	7 → ♭7	F major
C Dorian	C D E♭ F G A B♭	1 2 ♭3 4 5 6 ♭7	3 → ♭3	B♭ major
C Aeolian (natural minor)	C D E♭ F G A♭ B♭	1 2 ♭3 4 5 ♭6 ♭7	6 → ♭6	E♭ major
C Phrygian	C D♭ E♭ F G A♭ B♭	1 ♭2 ♭3 4 5 ♭6 ♭7	2 → ♭2	A♭ major
C Locrian	C D♭ E♭ F G♭ A♭ B♭	1 ♭2 ♭3 4 ♭5 ♭6 ♭7	5 → ♭5	D♭ major

Each step down lowers exactly one note. Major-type modes: Lydian, Ionian, Mixolydian. Minor-type: Dorian, Aeolian, Phrygian. Locrian is diminished. To find a mode's parent major scale: Dorian = major a whole step down, Phrygian = a major 3rd down, Lydian = a 4th down, Mixolydian = a 5th down, Aeolian = a minor 3rd up, Locrian = a half step up.

Minor keys: natural minor (Aeolian view) plus the two "fixes"

°	Natural minor (A)	Harmonic minor (A)	Melodic minor (A)	What players actually use
1	**i7** Am7	**i(maj7)** Am(maj7)	**i(maj7)** Am(maj7)	i7, i6, i(maj7)
2	**iiø7** Bm7♭5	**iiø7** Bm7♭5	**ii7** Bm7	iiø7
3	**♭IIImaj7** Cmaj7	**♭III+maj7** Cmaj7♯5	**♭III+maj7** Cmaj7♯5	♭IIImaj7
4	**iv7** Dm7	**iv7** Dm7	**IV7** D7	iv7 (iv6)
5	**v7** Em7	**V7** E7	**V7** E7	V7 (♭9) — from harmonic minor
6	**♭VImaj7** Fmaj7	**♭VImaj7** Fmaj7	**viø7** F♯m7♭5	♭VImaj7
7	**♭VII7** G7	**vii°7** G♯°7	**viiø7** G♯m7♭5	♭VII7 (or vii°7)

Roman numerals are always measured against the **major** scale of the tonic, so minor-key chords carry flats: i iiø ♭III iv v ♭VI ♭VII. Natural minor has a weak minor v; harmonic minor raises the 7th to make a real V7 (and vii°7); melodic minor also raises the 6th (smooth line, IV7, iiø with ♮9). In practice a minor key mixes all three.

Symmetric scales (repeat inside the octave, so each has only a few distinct transpositions)

Scale	Formula	On C	Pattern	Distinct	Use over	Contains
Half-whole diminished	1 ♭2 ♯2 3 ♯4 5 6 ♭7	C D♭ D♯ E F♯ G A B♭	H W H W…	3	7♭9, 13♭9, 7♯9 (dominant)	four dom7s (C7 E♭7 G♭7 A7), dim7s, the ♭9/♯9/♯11/13 tensions
Whole-half diminished	1 2 ♭3 4 ♭5 ♭6 𝄫7 7	C D E♭ F G♭ A♭ B𝄫 B	W H W H…	3	dim7 chords; °(maj7)	two dim7 chords a half step apart
Whole tone	1 2 3 ♯4 ♯5 ♭7	C D E F♯ G♯ B♭	all whole steps	2	7♯5, 9♯5, 7♭5	two augmented triads, 7♯5 chords
Augmented (hexatonic)	1 ♭3 3 5 ♯5 7	C E♭ E G G♯ B	m3 H m3 H…	4	maj7♯5, major-3rd cycles	two augmented triads, three major + three minor triads
Chromatic	1 ♭2 2 ♭3 3 4 ♯4 5 ♭6 6 ♭7 7	C D♭ D E♭ E F F♯ G A♭ A B♭ B	all half steps	1	passing tones only	everything

Half-whole from the root = whole-half from the ♭9 (or from the 3, 5, ♭7). Diminished shapes repeat every **3 frets**, whole-tone shapes every **2 frets**.

6 Chord–scale cheat sheet: what to play over what

By function (in C)

Function	Example	First choice	Alternatives
Imaj7 (tonic)	Cmaj7	**Ionian · major pentatonic**	Lydian, bebop major
IVmaj7	Fmaj7	**Lydian**	Major pentatonic on the 1 or 5
ii7	Dm7	**Dorian**	Minor pentatonic, bebop Dorian
iii7	Em7	**Phrygian (♭2 only as a passing note)**	Minor pentatonic
vi7	Am7	**Aeolian**	Dorian when it acts like a ii (vi–II7–ii–V)
V7 → I (major)	G7	**Mixolydian · bebop dominant**	Altered or half-whole for tension
V7 → i (minor)	G7♭9 → Cm	**Phrygian dominant (= harmonic minor of the target)**	Altered, half-whole diminished
V7alt	G7alt	**Altered (melodic minor a half step up)**	Minor pentatonic on the ♭3
iiø7 in minor	Dm7♭5	**Locrian ♮2 (melodic minor on the ♭3)**	Locrian
viiø7 in major	Bm7♭5	**Locrian**	—
Minor tonic (im6, im(maj7))	Cm6	**Melodic minor**	Dorian, harmonic minor
Non-resolving dominant (II7, ♭VII7, IV7 in minor)	D7, B♭7	**Lydian dominant**	Mixolydian
Tritone sub ♭II7	D♭7 → C	**Lydian dominant (same notes as G altered)**	—
Secondary dominant to a minor chord	A7 → Dm	**Mixolydian ♭6 / Phrygian dominant**	Altered
Secondary dominant to a major chord	D7 → G	**Mixolydian**	Lydian dominant
Passing dim7	C♯°7 → Dm7	**Whole-half diminished**	Harmonic minor of the target
7sus4	G7sus4	**Mixolydian (4 is now a chord tone)**	Minor pentatonic on the 5
Blues (I7 IV7 V7)	C7 F7 G7	**Blues scale / minor pentatonic of the key**	Major pentatonic, each chord's Mixolydian

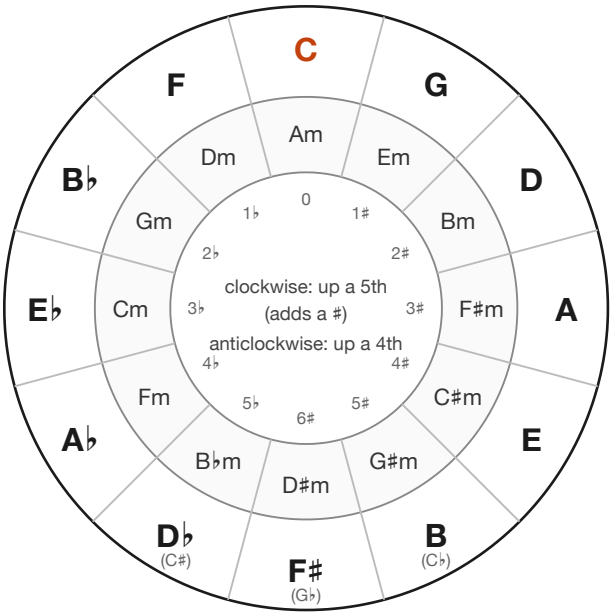
By chord quality: first choice, other options, and why

Chord	First choice	Other options	Why the first choice works
Major 7 (Imaj7, IVmaj7)	**Ionian (major)**	Major pentatonic · Lydian · Major pentatonic on the 5 · Minor pentatonic on the 3	Inside sound. Treat the 4th as a passing note (it's an avoid note over the 3rd).
Major 7 ♯11 (Lydian chord)	**Lydian**	Major pentatonic on the 2 · Minor pentatonic on the 7 · Major pentatonic on the 5 · Lydian #2	The chord is literally Lydian's tonic chord.
Major 6 (tonic)	**Major pentatonic**	Minor pentatonic on the 6 · Ionian (major) · Bebop major · Major pentatonic on the 5	C6 is four of its five notes; add the 9.
Dominant 7	**Mixolydian**	Major pentatonic · Blues scale · Minor pentatonic on the 5 · Dorian on the 5	Inside: 9 and 13. Watch the 4th (avoid note over the 3rd).
Dominant 7sus4 (9sus4, 13sus4)	**Mixolydian**	Minor pentatonic on the 5 · Major pentatonic on the 4 · Dorian on the 5 · Dorian ♭2	The 4th is now a chord tone, not an avoid note; avoid leaning on the 3rd.
Dominant 7♭9	**Phrygian dominant**	Harmonic minor on the 4 · Half-whole diminished · Whole-half diminished on the 3 · Altered	Default when resolving to minor: ♭9 and ♭13 are notes of the minor key.
Dominant 7♯9 ('Hendrix chord')	**Minor pentatonic**	Blues scale · Half-whole diminished · Altered · Minor pentatonic on the ♭3	Its ♭3 is the chord's ♯9: the rock/blues sound.
Altered dominant	**Altered**	Melodic minor on the ♭2 · Minor pentatonic on the ♭3 · Lydian dominant on the ♭5 · Minor 6 pentatonic on the ♭2	All tensions altered (♭9 ♯9 ♯11 ♭13), 3rd and ♭7 kept. G7alt = A♭ melodic minor.
Dominant 7♯11 (Lydian dominant)	**Lydian dominant**	Melodic minor on the 5 · Dominant pentatonic on the 2 · Whole tone · Altered on the ♭5	1 2 3 ♯4 5 6 ♭7: the chord-scale for 7♯11 (G7♯11 = D melodic minor).
Dominant 13	**Mixolydian**	Major pentatonic · Minor pentatonic on the 6 · Dorian on the 5 · Bebop dominant	Contains 9 and 13; the default.
Minor 7 (ii7, vi7, i7)	**Dorian**	Minor pentatonic · Ionian (major) on the ♭7 · Aeolian (natural minor) · Minor pentatonic on the 5	Default for ii7 and modal i7: natural 6 is the colour.
Minor 6 (tonic minor)	**Dorian**	Melodic minor · Minor 6 pentatonic · Minor pentatonic · Bebop melodic minor	Contains the 6; ♭7 as a passing note.
Minor-major 7	**Melodic minor**	Harmonic minor · Minor 6 pentatonic · Bebop melodic minor · Dominant pentatonic on the 5	The tonic chord of melodic minor.
Half-diminished (iiø7, viiø7)	**Locrian**	Locrian ♮2 · Melodic minor on the ♭3 · Minor 6 pentatonic on the ♭3 · Mixolydian on the ♭6	Inside for viiø7 in major; ♭9 sounds dark.
Diminished 7 (passing / vii°7)	**Whole-half diminished**	Harmonic minor on the ♭2 · Half-whole diminished on the ♭6 · Bebop major on the ♭2	Symmetric 8-note scale containing the dim7 plus a note a whole step above each chord tone.
Major 7 ♯5 (♭III+ of melodic minor)	**Lydian augmented**	Melodic minor on the 6 · Ionian ♯5 · Augmented (hexatonic)	1 2 3 ♯4 ♯5 6 7: the chord-scale. Cmaj7♯5 = A melodic minor.
Dominant 7♯5 (augmented 7)	**Whole tone**	Altered · Mixolydian ♭6	9, ♯11, ♯5: the dreamy augmented sound.
Dominant 7♭13	**Mixolydian ♭6**	Phrygian dominant · Altered · Whole tone	1 2 3 4 5 ♭6 ♭7: natural 9 with ♭13. G7♭13 = C melodic minor.

Chord	First choice	Other options	Why the first choice works
Dominant 7♭9b13 (minor-key V)	**Phrygian dominant**	Harmonic minor on the 4 · Altered	Exactly the chord's tensions: ♭9, ♭13, natural 11 as passing.
Minor blues (whole form, from the key root)	**Minor pentatonic**	Blues scale · Dorian · Aeolian (natural minor) · Melodic minor	Works over i7 and iv7; the basic sound.
Major / dominant blues (whole form, from the key root)	**Minor pentatonic**	Blues scale · Major pentatonic · Major blues scale · Mixolydian	Works over the whole form: the ♭3 and ♭7 are blue notes.

"On the ♭3" etc. means start that scale on that degree of the chord (G7alt: minor pentatonic on the ♭3 = B♭ minor pentatonic). Whatever the scale, land chord tones on strong beats.

7 Keys: circle of fifths, signatures, diatonic chords



Major	Rel. minor	♯	Accidentals
C	Am	—	—
G	Em	1 ♯	F♯
D	Bm	2 ♯	F♯ C♯
A	F♯m	3 ♯	C♯ F♯ G♯
E	C♯m	4 ♯	F♯ G♯ C♯ D♯
B	G♯m	5 ♯	C♯ D♯ F♯ G♯ A♯
F♯	D♯m	6 ♯	F♯ G♯ A♯ C♯ D♯ E♯
D♭	B♭m	5 ♭	D♭ E♭ G♭ A♭ B♭
A♭	Fm	4 ♭	A♭ B♭ D♭ E♭
E♭	Cm	3 ♭	E♭ A♭ B♭
B♭	Gm	2 ♭	B♭ E♭
F	Dm	1 ♭	B♭

Order of sharps: F C G D A E B. Order of flats: B E A D G C F. Relative minor = 6th degree (3 frets below the major root); same notes, same key signature.

Diatonic 7th chords in every major key (drop the 7th for the triad: I ii iii IV V vi vii°)

Key	Imaj7	ii7	iii7	IVmaj7	V7	vi7	viiø7
C	Cmaj7	Dm7	Em7	Fmaj7	G7	Am7	Bm7♭5
G	Gmaj7	Am7	Bm7	Cmaj7	D7	Em7	F♯m7♭5
D	Dmaj7	Em7	F♯m7	Gmaj7	A7	Bm7	C♯m7♭5
A	Amaj7	Bm7	C♯m7	Dmaj7	E7	F♯m7	G♯m7♭5
E	Emaj7	F♯m7	G♯m7	Amaj7	B7	C♯m7	D♯m7♭5
B	Bmaj7	C♯m7	D♯m7	Emaj7	F♯7	G♯m7	A♯m7♭5
F♯	F♯maj7	G♯m7	A♯m7	Bmaj7	C♯7	D♯m7	E♯m7♭5
G♭	G♭maj7	A♭m7	B♭m7	C♭maj7	D♭7	E♭m7	Fm7♭5
D♭	D♭maj7	E♭m7	Fm7	G♭maj7	A♭7	B♭m7	Cm7♭5
A♭	A♭maj7	B♭m7	Cm7	D♭maj7	E♭7	Fm7	Gm7♭5
E♭	E♭maj7	Fm7	Gm7	A♭maj7	B♭7	Cm7	Dm7♭5
B♭	B♭maj7	Cm7	Dm7	E♭maj7	F7	Gm7	Am7♭5
F	Fmaj7	Gm7	Am7	B♭maj7	C7	Dm7	Em7♭5

Diatonic 7th chords in every minor key (natural minor, plus the harmonic-minor V7)

Key	i7	iiø7	♭IIImaj7	iv7	v7	♭VImaj7	♭VII7	V7 (harm.)
Am	Am7	Bm7♭5	Cmaj7	Dm7	Em7	Fmaj7	G7	E7
Em	Em7	F♯m7♭5	Gmaj7	Am7	Bm7	Cmaj7	D7	B7
Bm	Bm7	C♯m7♭5	Dmaj7	Em7	F♯m7	Gmaj7	A7	F♯7
F♯m	F♯m7	G♯m7♭5	Amaj7	Bm7	C♯m7	Dmaj7	E7	C♯7
C♯m	C♯m7	D♯m7♭5	Emaj7	F♯m7	G♯m7	Amaj7	B7	G♯7
G♯m	G♯m7	A♯m7♭5	Bmaj7	C♯m7	D♯m7	Emaj7	F♯7	D♯7
D♯m	D♯m7	E♯m7♭5	F♯maj7	G♯m7	A♯m7	Bmaj7	C♯7	A♯7
E♭m	E♭m7	Fm7♭5	G♭maj7	A♭m7	B♭m7	C♭maj7	D♭7	B♭7
B♭m	B♭m7	Cm7♭5	D♭maj7	E♭m7	Fm7	G♭maj7	A♭7	F7
Fm	Fm7	Gm7♭5	A♭maj7	B♭m7	Cm7	D♭maj7	E♭7	C7
Cm	Cm7	Dm7♭5	E♭maj7	Fm7	Gm7	A♭maj7	B♭7	G7
Gm	Gm7	Am7♭5	B♭maj7	Cm7	Dm7	E♭maj7	F7	D7
Dm	Dm7	Em7♭5	Fmaj7	Gm7	Am7	B♭maj7	C7	A7

Functions: **tonic** I iii vi · **predominant** ii IV · **dominant** V vii°. Qualities never change: maj7 m7 m7 maj7 7 m7 m7♭5 in every major key.

8 Progressions and functional harmony

Roman numerals relative to the major scale of the tonic; examples in C (minor ones in C minor).

Functions

Function	Major key	Minor key	Feels
Tonic	I, iii, vi	i, ♭III, (♭VI)	home, stable
Predominant (subdominant)	ii, IV (and borrowed iv, iiø7, ♭VI)	iiø, iv, ♭VI	moving away
Dominant	V, vii°	V (from harmonic minor), vii°7, ♭VII	tension: the tritone between 3 and ♭7 wants to resolve

Cadences, turnarounds and stock progressions

Name	Roman	In C	Notes
Authentic cadence	**IV | V7 | I**	F | G7 | C	The strongest ending: V7 → I (the 3rd of V rises to the root, the 7th falls to the 3rd).
Plagal cadence	**IV | I**	F | C	"Amen" ending; softer than V → I.
Major ii–V–I	**ii7 | V7 | Imaj7**	Dm7 | G7 | Cmaj7	The core of jazz harmony. Learn it in all 12 keys, in every voicing type.
Minor ii–V–i	**iiø7 | V7♭9 | im6**	Dm7♭5 | G7♭9 | Cm6	iiø7 and V7♭9 from harmonic minor; the tonic is often m6 or m(maj7).
I–vi–ii–V turnaround	**Imaj7 vi7 | ii7 V7**	Cmaj7 Am7 | Dm7 G7	Two-bar loop at the end of a tune to return to the top.
iii–VI–ii–V	**iii7 VI7 | ii7 V7**	Em7 A7 | Dm7 G7	iii replaces I; VI7 is V7/ii (secondary dominant).
I–VI7–ii–V	**Imaj7 VI7 | ii7 V7**	Cmaj7 A7 | Dm7 G7	Make vi dominant for a stronger pull to ii.
Tritone turnaround	**Imaj7 ♭III7 | ♭VI7 ♭II7**	Cmaj7 E♭7 | A♭7 D♭7	Every dominant of I–VI–II–V replaced by its tritone sub.
Dameron turnaround	**Imaj7 ♭IIImaj7 | ♭VImaj7 ♭IImaj7**	Cmaj7 E♭maj7 | A♭maj7 D♭maj7	Same roots as above, all maj7: a borrowed-chord chain sliding home.
Backdoor ii–V	**iv7 ♭VII7 | Imaj7**	Fm7 B♭7 | Cmaj7	♭VII7 resolves up a whole step: softer, from the parallel minor.
Pop / axis	**I | V | vi | IV**	C | G | Am | F	Rotate it: vi–IV–I–V and IV–I–V–vi are the same loop.
50s / doo-wop	**I | vi | IV | V**	C | Am | F | G	Same as I–vi–ii–V with IV in place of ii.
Andalusian cadence	**i | ♭VII | ♭VI | V**	Cm | B♭ | A♭ | G	Minor key; bass walks 1–♭7–♭6–5, V is major (harmonic minor).
Pachelbel / Canon	**I V | vi iii | IV I | IV V**	C G | Am Em | F C | F G	Bass falls by step; endless melodic options.
Mixolydian rock	**I | ♭VII | IV | I**	C | B♭ | F | C	♭VII from Mixolydian; the ♭7 is the colour.
Minor rock	**i | ♭VI | ♭VII | i**	Cm | A♭ | B♭ | Cm	Aeolian; ♭VI–♭VII–i is the minor "power cadence".
Passing diminished	**Imaj7 #i°7 | ii7 #ii°7 | iii7**	Cmaj7 C#°7 | Dm7 D#°7 | Em7	A dim7 a half step below a chord = its rootless V7♭9.
Cycle of 4ths (diatonic)	**Imaj7 IVmaj7 | viiø7 iii7 | vi7 ii7 | V7 Imaj7**	Cmaj7 Fmaj7 | Bm7♭5 Em7 | Am7 Dm7 | G7 Cmaj7	Every root moves up a 4th: the strongest diatonic motion; ideal for voice-leading drills.
Coltrane cycle	**Imaj7 ♭III7 | ♭VImaj7 VII7 | IIImaj7 V7 | Imaj7**	Cmaj7 E♭7 | A♭maj7 B7 | Emaj7 G7 | Cmaj7	Three tonics a major 3rd apart, each reached by its V7.

12-bar blues (dominant)

I7 C7	**IV7** F7	**I7** C7	**I7** C7
IV7 F7	**IV7** F7	**I7** C7	**I7** C7
V7 G7	**IV7** F7	**I7** C7	**V7** G7

Bar 2 = "quick change" to IV (optional). Bars 11–12 are the turnaround.

Jazz blues

I7 C7	**IV7** F7	**I7** C7	**v7 I7** Gm7 C7
IV7 F7	**#iv°7** F#°7	**I7** C7	**iii7 VI7** Em7 A7
ii7 Dm7	**V7** G7	**I7 VI7** C7 A7	**ii7 V7** Dm7 G7

Minor blues

i7 Cm7	**iv7** Fm7	**i7** Cm7	**i7** Cm7
iv7 Fm7	**iv7** Fm7	**i7** Cm7	**i7** Cm7
♭VI7 A♭7	**V7** G7	**i7** Cm7	**V7** G7

Rhythm changes — A section (AABA; bridge: III7 | % | VI7 | % | II7 | % | V7 | % |)

I6 vi7 C6 Am7	**ii7 V7** Dm7 G7	**iii7 vi7** Em7 Am7	**ii7 V7** Dm7 G7
Imaj7 I7 Cmaj7 C7	**IVmaj7 ♯iv°7** Fmaj7 F♯°7	**I6/5 V7** C6/G G7	**I6** C6

Bridge in C: E7 | E7 | A7 | A7 | D7 | D7 | G7 | G7 — a chain of dominants around the cycle of 4ths.

Secondary dominants: V7 of each diatonic chord (in C)

Symbol	Chord → target	Chromatic note	Its own ii–V	Scale
V7/ii = VI7	**A7** → Dm7	C♯ (its 3rd, the leading tone)	Em7♭5 – A7♭9	Mixolydian ♭6 / Phrygian dominant (♭9 ♭13)
V7/iii = VII7	**B7** → Em7	D♯ (its 3rd, the leading tone)	F♯m7♭5 – B7♭9	Mixolydian ♭6 / Phrygian dominant (♭9 ♭13)
V7/IV = I7	**C7** → Fmaj7	E (its 3rd, the leading tone)	Gm7 – C7	Mixolydian (V7/V: Lydian dominant also good)
V7/V = II7	**D7** → G7	F♯ (its 3rd, the leading tone)	Am7 – D7	Mixolydian (V7/V: Lydian dominant also good)
V7/vi = III7	**E7** → Am7	G♯ (its 3rd, the leading tone)	Bm7♭5 – E7♭9	Mixolydian ♭6 / Phrygian dominant (♭9 ♭13)

Tritone substitution: dominants a ♭5 apart share their 3rd and ♭7

Pair	Shared tritone	How
G7 ↔ D♭7	B + F	B is the ♭7 of D♭7 (spelled C♭); F is its 3rd
C7 ↔ G♭7	E + B♭	E is the ♭7 of G♭7 (spelled F♭); B♭ is its 3rd
D7 ↔ A♭7	F♯ + C	F♯ is the ♭7 of A♭7 (spelled G♭); C is its 3rd
A7 ↔ E♭7	C♯ + G	C♯ is the ♭7 of E♭7 (spelled D♭); G is its 3rd
E7 ↔ B♭7	G♯ + D	G♯ is the ♭7 of B♭7 (spelled A♭); D is its 3rd
B7 ↔ F7	D♯ + A	D♯ is the ♭7 of F7 (spelled E♭); A is its 3rd

So D♭7 can replace G7 before C: the bass falls by half step (D♭ → C) and the guide tones are identical. Over the sub, play Lydian dominant; it equals the altered scale of the original V.

9 Modal interchange: borrowed chords

Modal interchange = keep the tonic, borrow chords from a **parallel** scale (same root, different mode). Each row below shows the 7th chords of C in that mode; "=" means identical to the C major chord. Numerals are against C major, so a borrowed chord carries its accidental (♭III, iv, ♭VI, ♭VII).

Mode on C	I	II	III	IV	V	VI	VII
Ionian (reference)	**Imaj7** Cmaj7	**ii7** Dm7	**iii7** Em7	**IVmaj7** Fmaj7	**V7** G7	**vi7** Am7	**viiø7** Bm7♭5
Lydian	=	**II7** D7	=	**♯ivø7** F♯m7♭5	**Vmaj7** Gmaj7	=	**vii7** Bm7
Mixolydian	**I7** C7	=	**iiiø7** Em7♭5	=	**v7** Gm7	=	**♭VIImaj7** B♭maj7
Dorian	**i7** Cm7	=	**♭IIImaj7** E♭maj7	**IV7** F7	**v7** Gm7	**viø7** Am7♭5	**♭VIImaj7** B♭maj7
Aeolian	**i7** Cm7	**iiø7** Dm7♭5	**♭IIImaj7** E♭maj7	**iv7** Fm7	**v7** Gm7	**♭VImaj7** A♭maj7	**♭VII7** B♭7
Phrygian	**i7** Cm7	**♭IImaj7** D♭maj7	**♭III7** E♭7	**iv7** Fm7	**vø7** Gm7♭5	**♭VImaj7** A♭maj7	**♭vii7** B♭m7
Locrian	**iø7** Cm7♭5	**♭IImaj7** D♭maj7	**♭iii7** E♭m7	**iv7** Fm7	**♭Vmaj7** G♭maj7	**♭VI7** A♭7	**♭vii7** B♭m7
Harmonic minor	**i(maj7)** Cm(maj7)	**iiø7** Dm7♭5	**♭III+maj7** E♭maj7♯5	**iv7** Fm7	=	**♭VImaj7** A♭maj7	**vii°7** B°7
Melodic minor	**i(maj7)** Cm(maj7)	=	**♭III+maj7** E♭maj7♯5	**IV7** F7	=	**viø7** Am7♭5	=
Harmonic major	=	**iiø7** Dm7♭5	=	**iv(maj7)** Fm(maj7)	=	**♭VI+maj7** A♭maj7♯5	**vii°7** B°7

The 15 moves you will meet most, in context

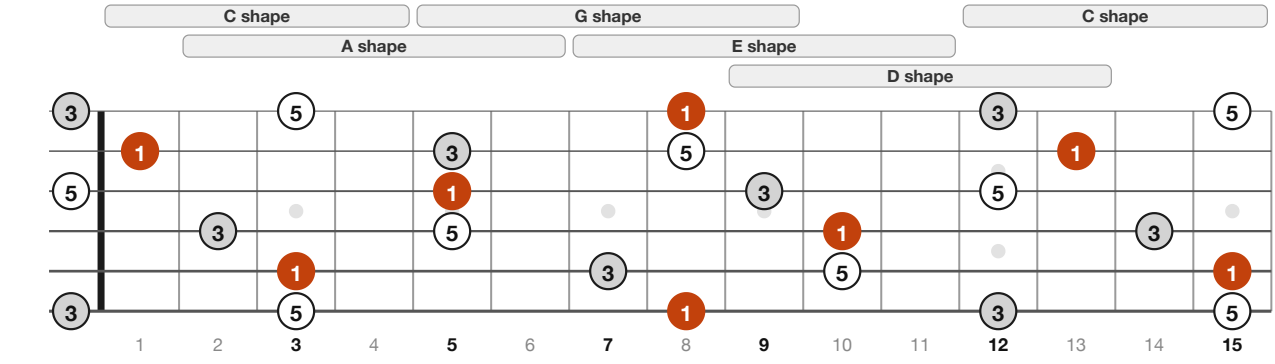
Move	Roman	In C	Borrowed from	What it does
IV → iv → I	**IVmaj7 | ivm6 | Imaj7**	Fmaj7 | Fm6 | Cmaj7	Aeolian	The ♭6 (A♭) sighs down to the 5th. The most-used borrowed chord.
iv7 → ♭VII7 → I (backdoor)	**iv7 | ♭VII7 | Imaj7**	Fm7 | B♭7 | Cmaj7	Aeolian	A soft dominant: ♭VII7 resolves up a whole step.
iiø7 → V7 → I	**iiø7 | V7♭9 | Imaj7**	Dm7♭5 | G7♭9 | Cmaj7	Aeolian / harmonic minor	Minor ii–V landing on a major tonic: dark approach, bright arrival.
V7 → ♭VImaj7 (deceptive) → V7 → I	**V7 | ♭VImaj7 | V7 | Imaj7**	G7 | A♭maj7 | G7 | Cmaj7	Aeolian	V7 dodges to ♭VI (leading tone holds as maj7), then tries again.
♭VI → ♭VII → I	**♭VI | ♭VII | I**	A♭ | B♭ | C	Aeolian	Rising "victory" cadence; no leading tone needed.
I → ♭VII → IV → I	**I | ♭VII | IV | I**	C | B♭ | F | C	Mixolydian	Rock staple; the ♭7 of the key in the bass.
I → v → IV	**I | v | IV | I**	C | Gm | F | C	Mixolydian / Dorian	Minor v: a "dominant" with no leading tone, laid-back.
I → ♭III → IV	**I | ♭III | IV | I**	C | E♭ | F | C	Aeolian / Dorian	Blues-rock colour: the ♭3 of the key as a major chord.
I → ♭IImaj7 → I	**Imaj7 | ♭IImaj7 | Imaj7**	Cmaj7 | D♭maj7 | Cmaj7	Phrygian	Half-step slide into the tonic; the Neapolitan colour.
I → II → IV → I	**I | II | IV | I**	C | D | F | C	Lydian	Major II (♯4 as its 3rd): bright, floating lift.
Imaj7 → i7 (parallel minor tonic)	**Imaj7 | % | i7 | %**	Cmaj7 | Cmaj7 | Cm7 | Cm7	Dorian / Aeolian	Same root, 3rd drops: instant darkening; common in bridges.
I → iv6 → I	**Imaj7 | ivm6 | Imaj7 | %**	Cmaj7 | Fm6 | Cmaj7 | Cmaj7	Melodic/Aeolian	iv6 = iiø7 with the 4th in the bass; the ♭6 → 5 and 6 → 5 voices.
vii°7 → I	**Imaj7 | vii°7 | Imaj7 | %**	Cmaj7 | B°7 | Cmaj7 | Cmaj7	Harmonic major / minor	The ♭6 turns viiø7 into a full dim7 leading-tone chord.
Dameron turnaround	**Imaj7 ♭IIImaj7 | ♭VImaj7 ♭IImaj7 | Imaj7**	Cmaj7 E♭maj7 | A♭maj7 D♭maj7 | Cmaj7	Aeolian + Phrygian	Chain of borrowed maj7 chords sliding back home.
Picardy third	**iv7 | V7 | I**	Fm7 | G7 | C	Ionian into a minor key	A minor-key piece ends on a major tonic chord.

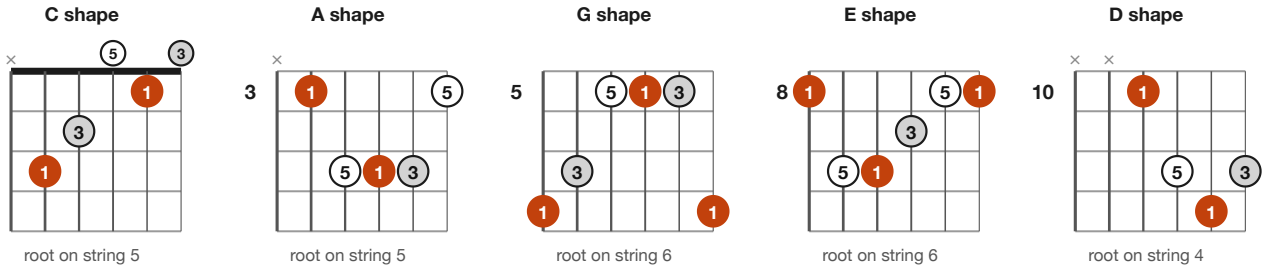
How to use it

- Find the major-key chord on the same degree, then compare: usually **one note** moves by a half step (A → A♭ for iv, ♭VI; B → B♭ for ♭VII). That note is the sound.
- Borrowed chords are mostly **predominants** (iv, iiø7, ♭VI, ♭VII7): they lead to V or straight to I.
- Over a borrowed chord, play the **source mode on the tonic** (C Aeolian notes over A♭maj7 and Fm7), or the chord's own mode (A♭ Lydian = C Aeolian).
- Darkness scale: Lydian (brightest) → Ionian → Mixolydian → Dorian → Aeolian → Phrygian → Locrian. Borrowing from further down = darker colour.

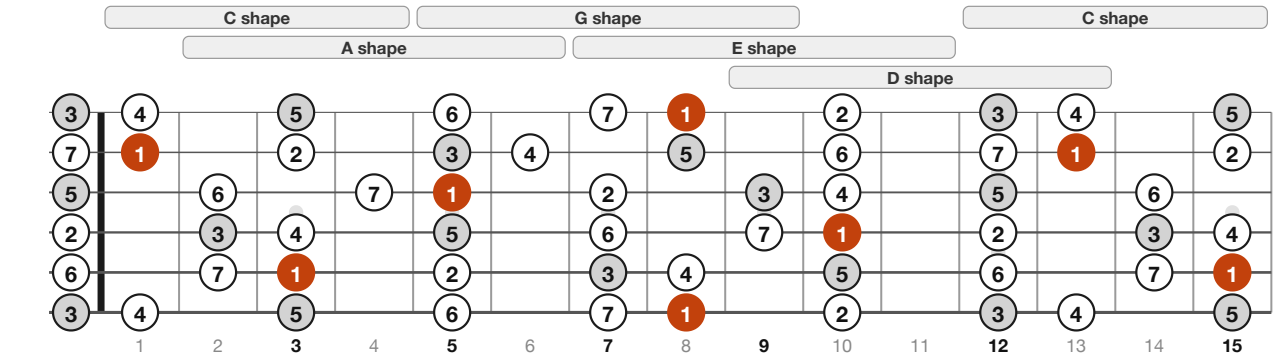
10 CAGED: one key across the whole neck

C major triad (1 3 5) over the whole neck, with the five CAGED windows





The key of C major over the whole neck: every scale note, chord tones of C shaded



Shape	Roots on	C chord: root fret	Scale window (C major)	Next shape up the neck
C	strings 5 and 2	3 (string 5)	frets 0–4	A
A	strings 5 and 3	3 (string 5)	frets 2–6	G
G	strings 6, 3 and 1	8 (string 6)	frets 5–9	E
E	strings 6, 4 and 1	8 (string 6)	frets 7–11	D
D	strings 4 and 2	10 (string 4)	frets 9–13	C

- The five open-chord shapes C, A, G, E, D, moved up the neck, tile the fretboard in that order and then repeat (C again at fret 12).
- Each chord shape carries a **scale shape**, an **arpeggio shape** and a **pentatonic box** with it (Part II). Learn them as one unit: chord → arpeggio → scale in the same window.
- Any key: find the root on string 6, 5 or 4, choose the shape whose root sits there, and the whole system shifts with it. The sequence is always C→A→G→E→D going up.

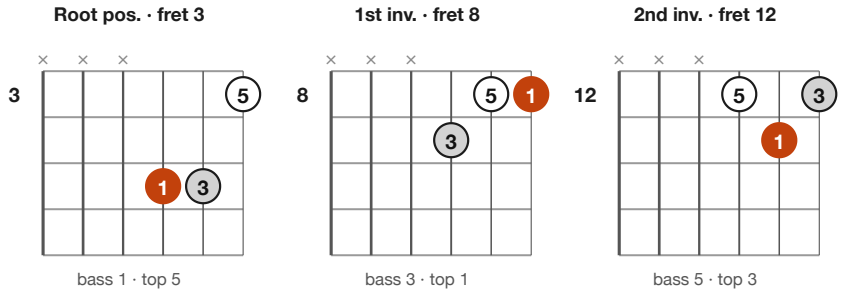
11 Triads in every inversion

Close-position triads (all three notes within an octave) on the four adjacent string sets, root C. The three inversions on one string set follow each other up the neck; each grip is movable — slide it so the accent dot lands on your root. The B string makes 3-2-1 and 4-3-2 shapes differ from 5-4-3 and 6-5-4.

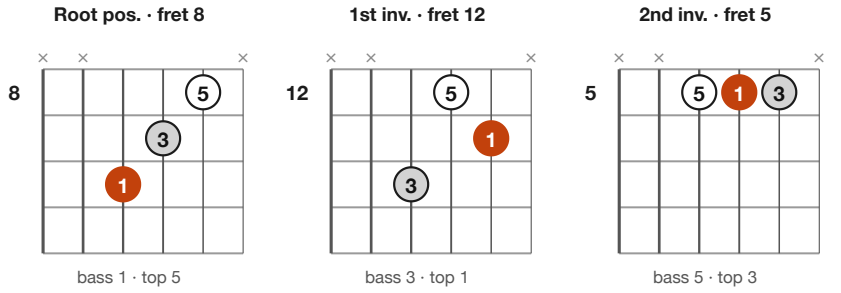
Major triad · 1 3 5 (C E G)

Tops: root position has the 5 on top, 1st inversion the 1, 2nd inversion the 3. Pick the inversion by the melody note you want on top.

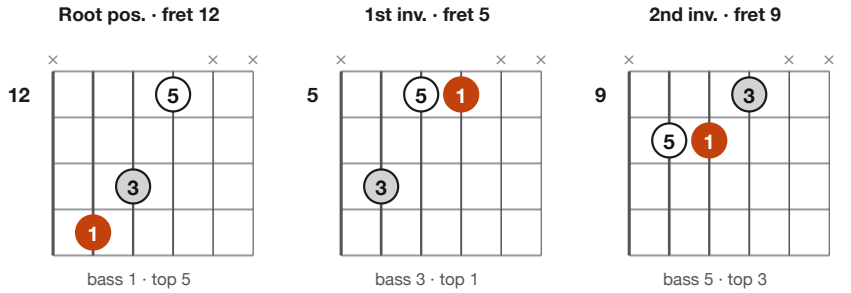
C
strings 3-2-1



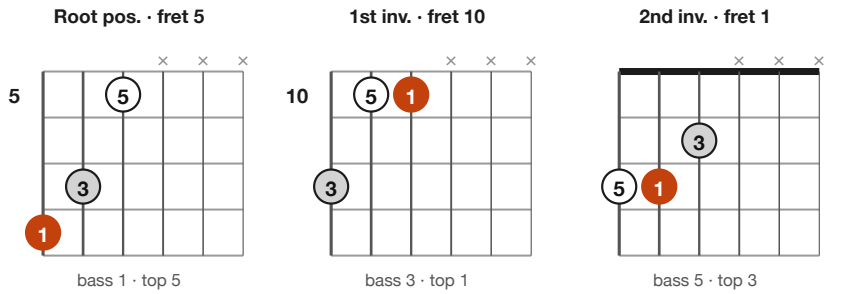
C
strings 4-3-2



C
strings 5-4-3



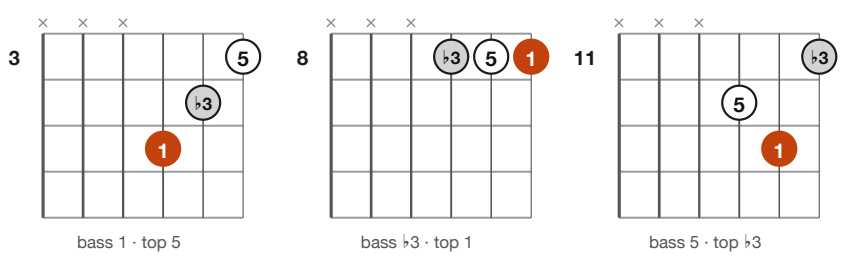
C
strings 6-5-4



Minor triad · 1 ♭3 5 (C E♭ G)

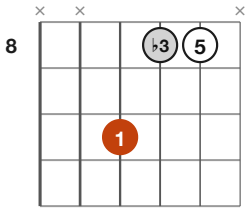
Every major shape with the 3 lowered one fret. Minor triads = ii, iii, vi of a major key.

Cm
strings 3-2-1



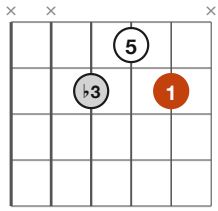
Cm
strings 4-3-2

Root pos. · fret 8



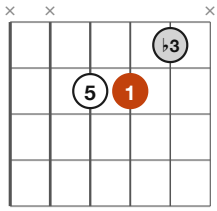
bass 1 · top 5

1st inv. · fret 12



bass ♭3 · top 1

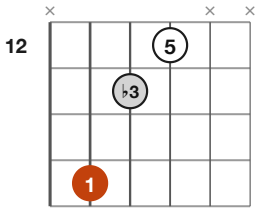
2nd inv. · fret 4



bass 5 · top ♭3

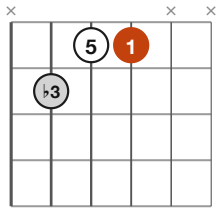
Cm
strings 5-4-3

Root pos. · fret 12



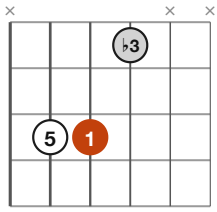
bass 1 · top 5

1st inv. · fret 5



bass ♭3 · top 1

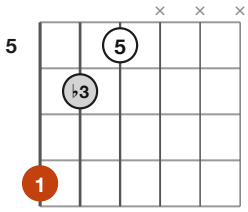
2nd inv. · fret 8



bass 5 · top ♭3

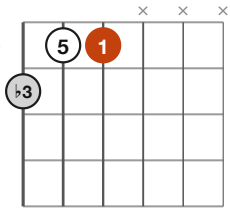
Cm
strings 6-5-4

Root pos. · fret 5



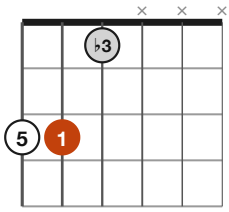
bass 1 · top 5

1st inv. · fret 10



bass ♭3 · top 1

2nd inv. · fret 1



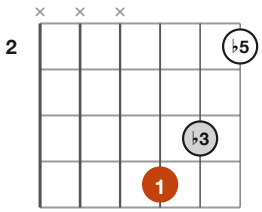
bass 5 · top ♭3

Diminished triad · 1 ♭3 ♭5 (C E♭ G♭)

Every minor shape with the 5 lowered one fret. vii° of a major key = the top of a V7 (rootless).

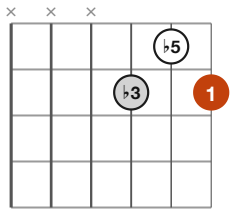
C°
strings 3-2-1

Root pos. · fret 2



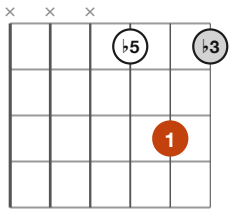
bass 1 · top ♭5

1st inv. · fret 7



bass ♭3 · top 1

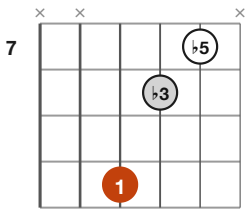
2nd inv. · fret 11



bass ♭5 · top ♭3

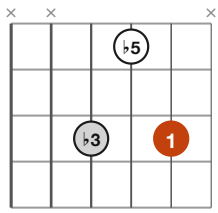
C°
strings 4-3-2

Root pos. · fret 7



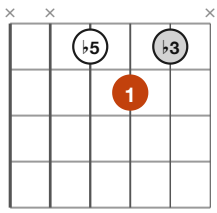
bass 1 · top ♭5

1st inv. · fret 11



bass ♭3 · top 1

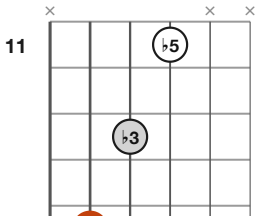
2nd inv. · fret 4



bass ♭5 · top ♭3

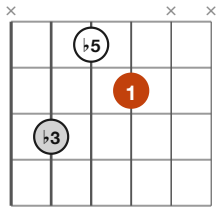
C°
strings 5-4-3

Root pos. · fret 11



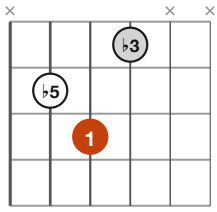
bass 1 · top ♭5

1st inv. · fret 4

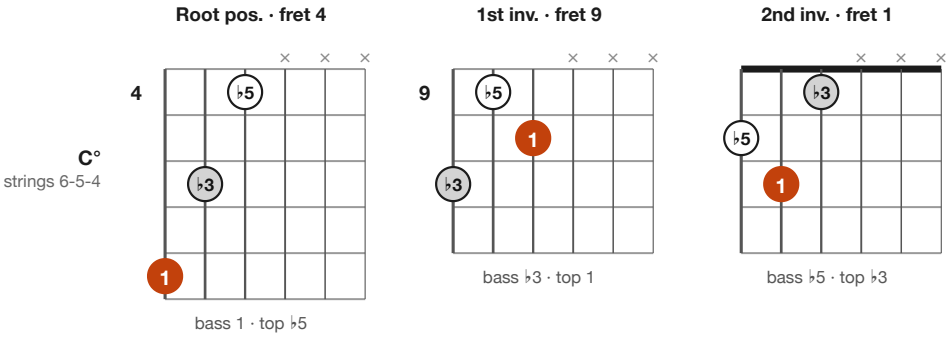


bass ♭3 · top 1

2nd inv. · fret 8

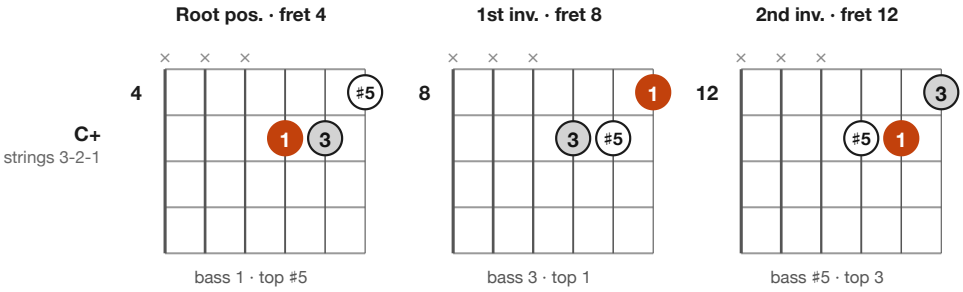


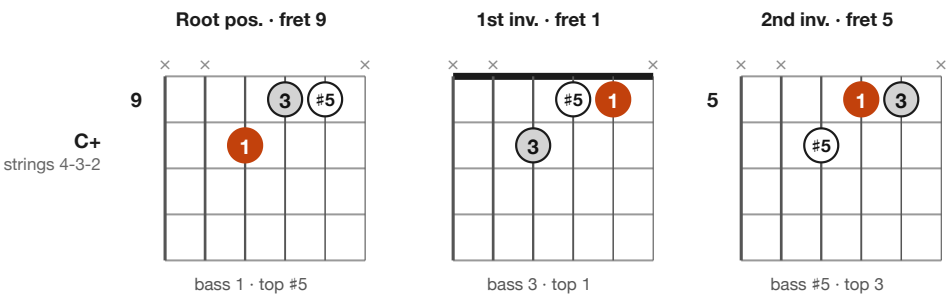
bass ♭5 · top ♭3

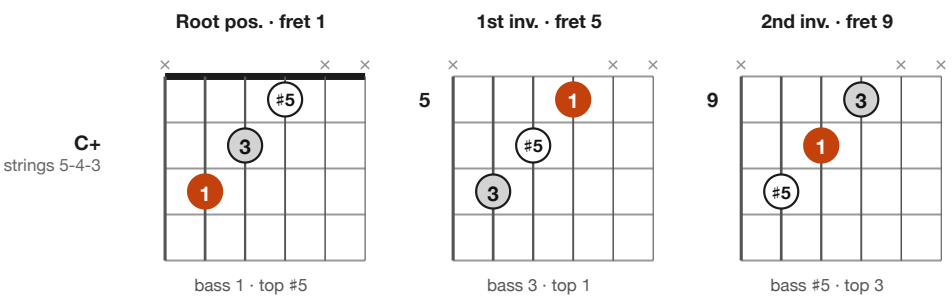


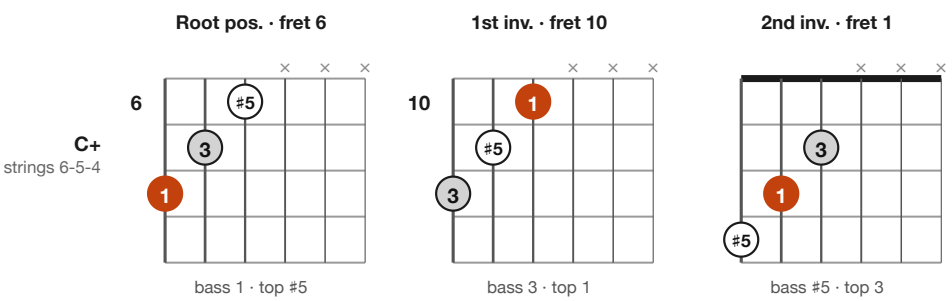
Augmented triad · 1 3 ♯5 (C E G♯)

Symmetric: every inversion is the same shape 4 frets higher, and each one is also the augmented triad on its own bass note.









Drills

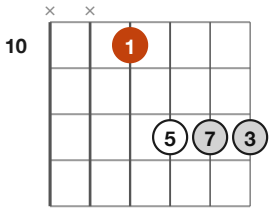
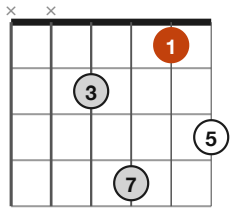
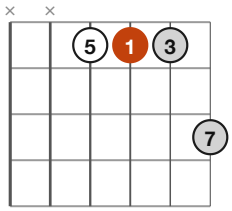
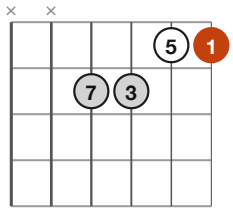
- One string set at a time: root position → 1st → 2nd → root position an octave up, saying the bass note interval aloud.
- Same chord, all four string sets, staying in one 5-fret area (frets 5–9): find the inversion that lives there on each set.
- Diatonic triads of C (C Dm Em F G Am B°) up one string set in one inversion, then in the inversion nearest the previous chord (voice leading).
- I–IV–V–I and ii–V–I with the smallest possible movement between chords; each voice moves by step or stays.

12 Drop 2 voicings

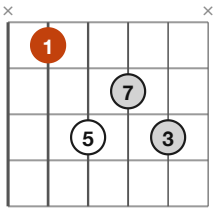
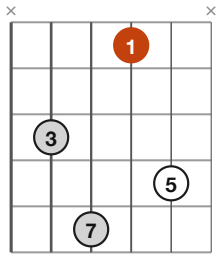
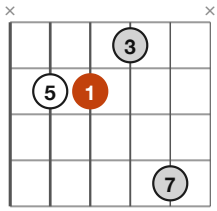
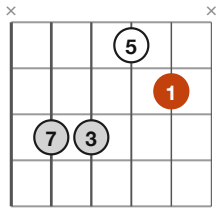
The workhorse jazz/comping voicing. All four inversions of each quality on strings 4-3-2-1 and 5-4-3-2 (and 6-5-4-3 for the three main qualities). Root C; grey = 3rd and 7th.

How drop 2 is built: take a close 7th chord (four notes inside an octave) and drop the **second-highest** note an octave. Root position C E G B → drop the G → **C G B E** (1 5 7 3). The result fits four adjacent strings with a one-finger-per-string grip. Each inversion puts a different chord tone in the bass (R → 3 → 5 → 7) and on top. Captions give the bass and top intervals: choose by the bass line you want or by the melody note on top.

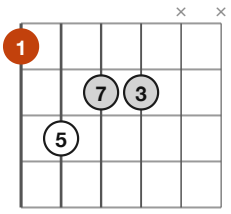
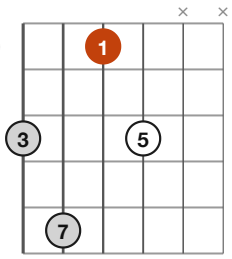
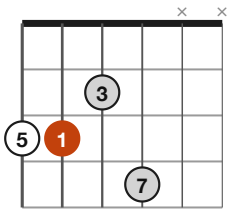
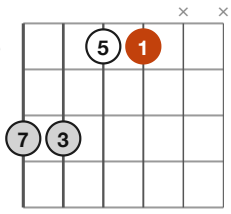
Cmaj7
strings 4-3-2-1

Root pos. · fret 10	1st inv. · fret 1	2nd inv. · fret 5	3rd inv. · fret 8
			
bass 1 · top 3	bass 3 · top 5	bass 5 · top 7	bass 7 · top 1

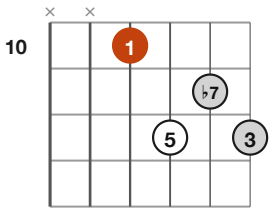
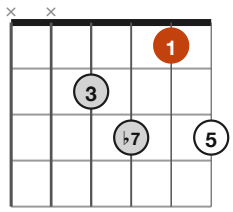
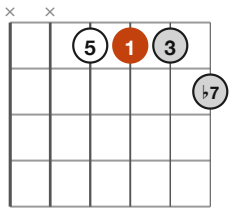
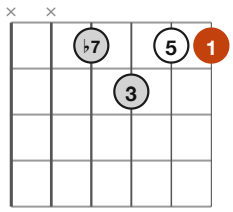
Cmaj7
strings 5-4-3-2

Root pos. · fret 3	1st inv. · fret 5	2nd inv. · fret 9	3rd inv. · fret 12
			
bass 1 · top 3	bass 3 · top 5	bass 5 · top 7	bass 7 · top 1

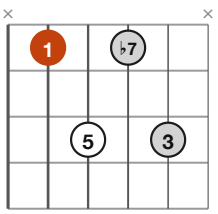
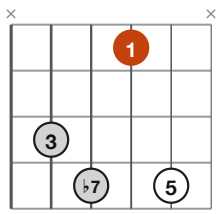
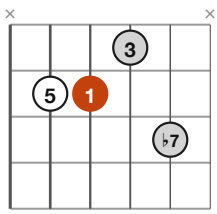
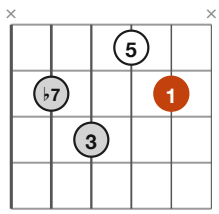
Cmaj7
strings 6-5-4-3

Root pos. · fret 8	1st inv. · fret 10	2nd inv. · fret 1	3rd inv. · fret 5
			
bass 1 · top 3	bass 3 · top 5	bass 5 · top 7	bass 7 · top 1

C7
strings 4-3-2-1

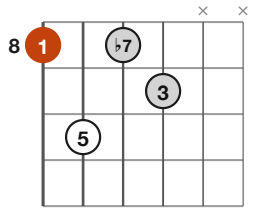
Root pos. · fret 10	1st inv. · fret 1	2nd inv. · fret 5	3rd inv. · fret 8
			
bass 1 · top 3	bass 3 · top 5	bass 5 · top ♭7	bass ♭7 · top 1

C7
strings 5-4-3-2

Root pos. · fret 3	1st inv. · fret 5	2nd inv. · fret 9	3rd inv. · fret 12
			
bass 1 · top 3	bass 3 · top 5	bass 5 · top ♭7	bass ♭7 · top 1

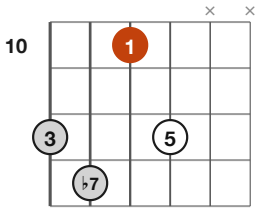
C7
strings 6-5-4-3

Root pos. · fret 8



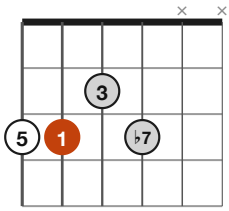
bass 1 · top 3

1st inv. · fret 10



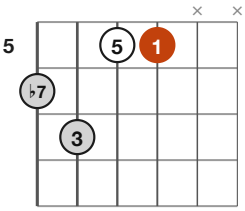
bass 3 · top 5

2nd inv. · fret 1



bass 5 · top ♭7

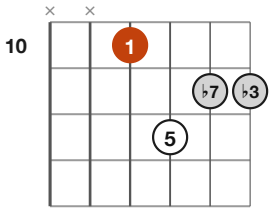
3rd inv. · fret 5



bass ♭7 · top 1

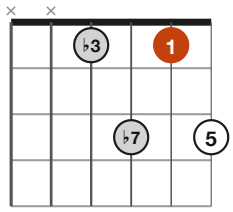
Cm7
strings 4-3-2-1

Root pos. · fret 10



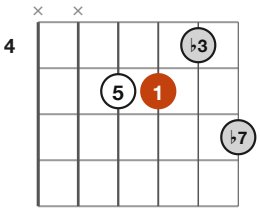
bass 1 · top ♭3

1st inv. · fret 1



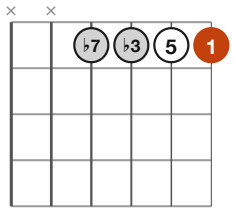
bass ♭3 · top 5

2nd inv. · fret 4



bass 5 · top ♭7

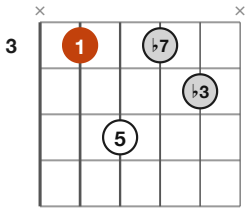
3rd inv. · fret 8



bass ♭7 · top 1

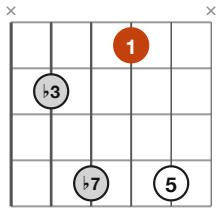
Cm7
strings 5-4-3-2

Root pos. · fret 3



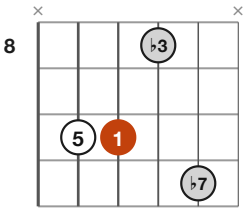
bass 1 · top ♭3

1st inv. · fret 5



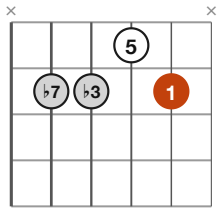
bass ♭3 · top 5

2nd inv. · fret 8



bass 5 · top ♭7

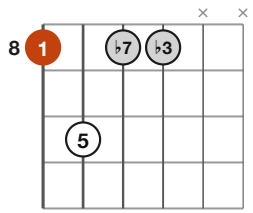
3rd inv. · fret 12



bass ♭7 · top 1

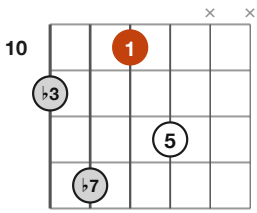
Cm7
strings 6-5-4-3

Root pos. · fret 8



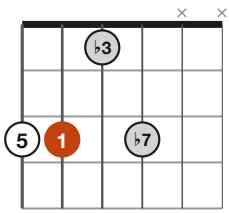
bass 1 · top ♭3

1st inv. · fret 10



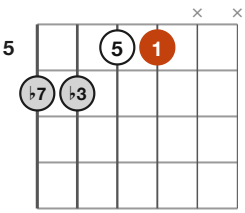
bass ♭3 · top 5

2nd inv. · fret 1



bass 5 · top ♭7

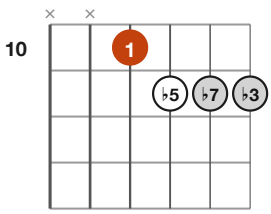
3rd inv. · fret 5



bass ♭7 · top 1

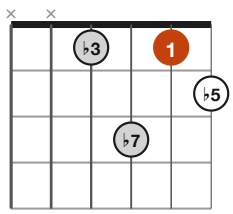
Cm7♭5
strings 4-3-2-1

Root pos. · fret 10



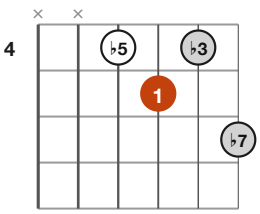
bass 1 · top ♭3

1st inv. · fret 1



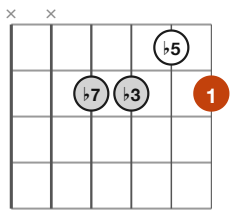
bass ♭3 · top ♭5

2nd inv. · fret 4



bass ♭5 · top ♭7

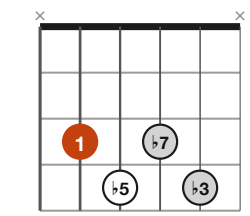
3rd inv. · fret 7



bass ♭7 · top 1

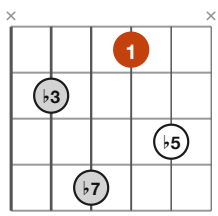
Cm7♭5
strings 5-4-3-2

Root pos. · fret 1



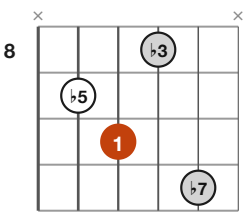
bass 1 · top ♭3

1st inv. · fret 5



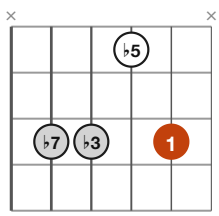
bass ♭3 · top ♭5

2nd inv. · fret 8



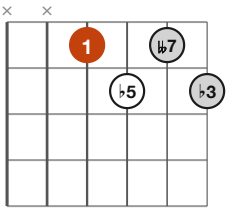
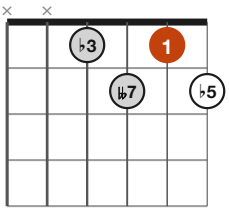
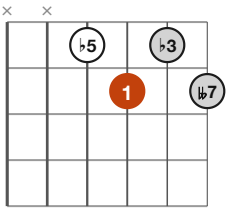
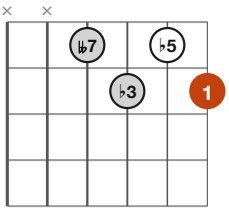
bass ♭5 · top ♭7

3rd inv. · fret 11

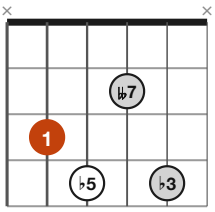
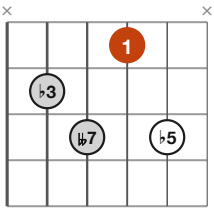
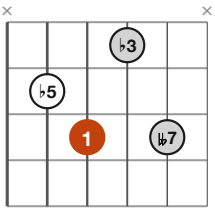
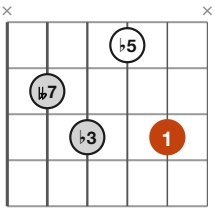


bass ♭7 · top 1

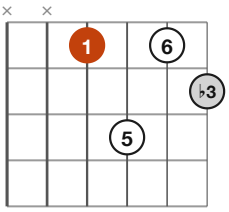
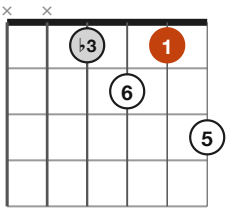
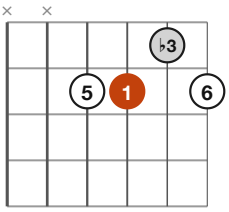
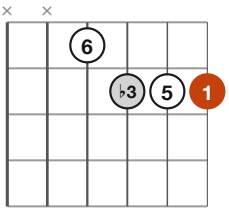
C°7
strings 4-3-2-1

Root pos. · fret 10	1st inv. · fret 1	2nd inv. · fret 4	3rd inv. · fret 7
10	4	7	
			
bass 1 · top ♭3	bass ♭3 · top ♭5	bass ♭5 · top 𝄫7	bass 𝄫7 · top 1

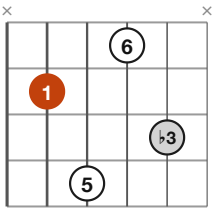
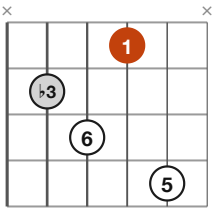
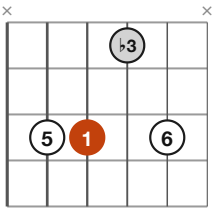
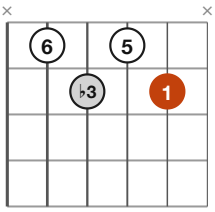
C°7
strings 5-4-3-2

Root pos. · fret 1	1st inv. · fret 5	2nd inv. · fret 8	3rd inv. · fret 11
	5	8	11
			
bass 1 · top ♭3	bass ♭3 · top ♭5	bass ♭5 · top 𝄫7	bass 𝄫7 · top 1

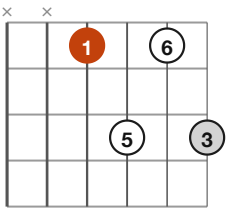
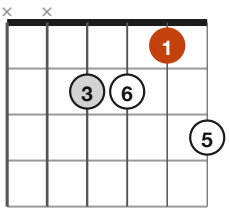
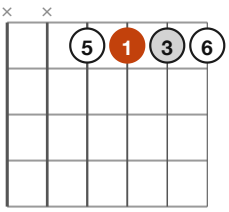
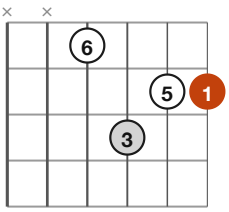
Cm6
strings 4-3-2-1

Root pos. · fret 10	1st inv. · fret 1	2nd inv. · fret 4	3rd inv. · fret 7
10	4	7	
			
bass 1 · top ♭3	bass ♭3 · top 5	bass 5 · top 6	bass 6 · top 1

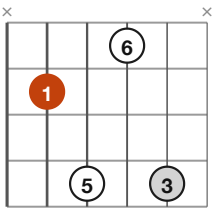
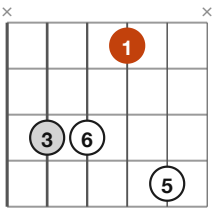
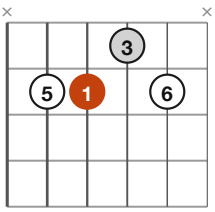
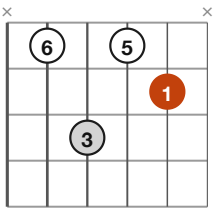
Cm6
strings 5-4-3-2

Root pos. · fret 2	1st inv. · fret 5	2nd inv. · fret 8	3rd inv. · fret 12
2	5	8	12
			
bass 1 · top ♭3	bass ♭3 · top 5	bass 5 · top 6	bass 6 · top 1

C6
strings 4-3-2-1

Root pos. · fret 10	1st inv. · fret 1	2nd inv. · fret 5	3rd inv. · fret 7
10	5	7	
			
bass 1 · top 3	bass 3 · top 5	bass 5 · top 6	bass 6 · top 1

C6
strings 5-4-3-2

Root pos. · fret 2	1st inv. · fret 5	2nd inv. · fret 9	3rd inv. · fret 12
2	5	9	12
			
bass 1 · top 3	bass 3 · top 5	bass 5 · top 6	bass 6 · top 1

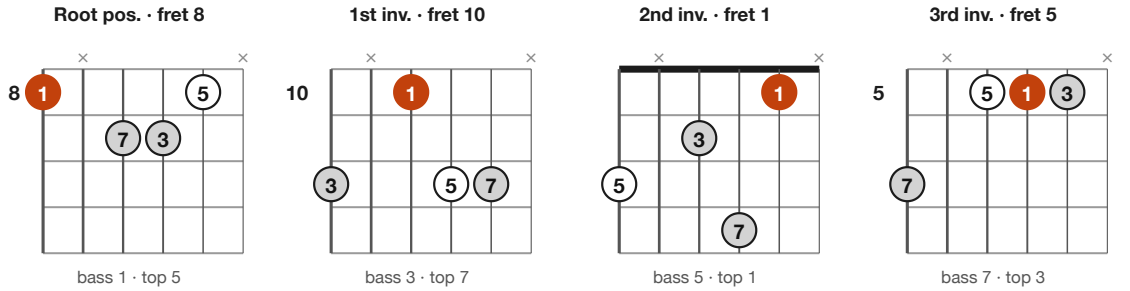
Shortcuts: from maj7, lower the 7 one fret → dominant 7; lower the 3 as well → m7; lower the 5 as well → m7♭5; lower the ♭7 again → °7. **°7** is symmetric (all four inversions are the same shape, 3 frets apart). **m6 and m7♭5 share notes:** Cm6 = Am7♭5 (C E♭ G A). **C6 = Am7.**

13 Drop 3 voicings

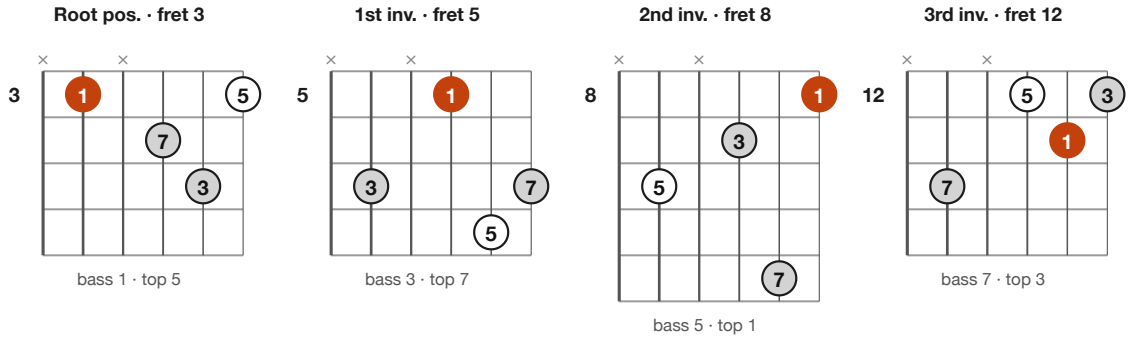
Bass note on string 6 or 5, one string skipped, three upper voices on adjacent strings. All four inversions for maj7, 7, m7 and m7♭5. Root C.

How drop 3 is built: drop the **third-highest** note of a close 7th chord an octave. From the close voicing B C E G (7 1 3 5), dropping the C gives **C B E G = 1 7 3 5**: root on string 6 or 5, one string skipped, then 7, 3, 5 on the next three strings — the classic jazz "barre" grip. The other three inversions put the 3, 5 or 7 in the bass.

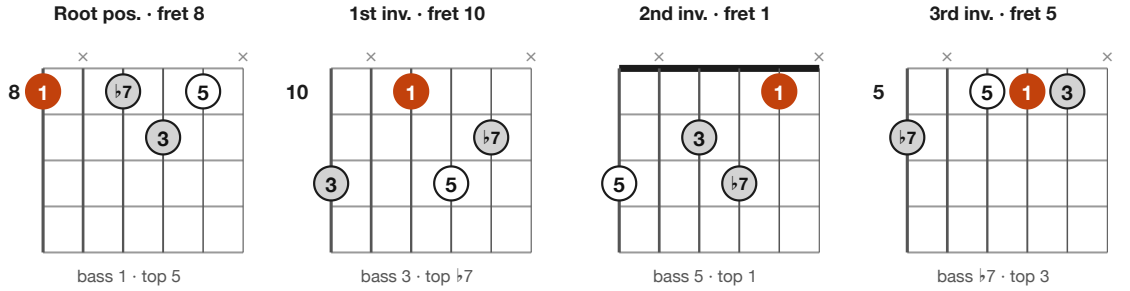
Cmaj7
strings 6-4-3-2



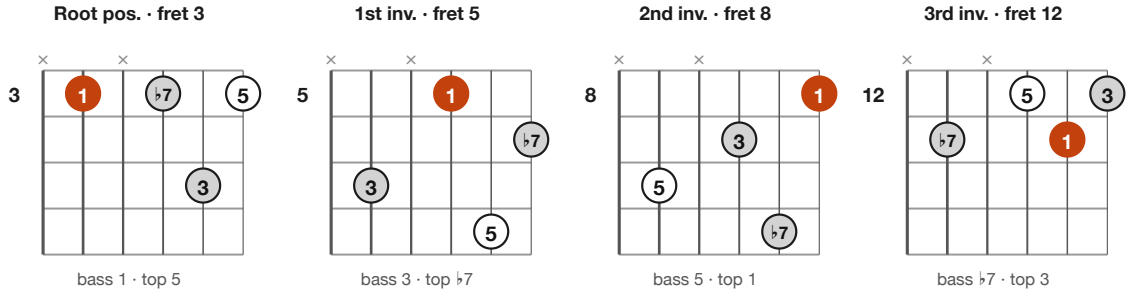
Cmaj7
strings 5-3-2-1



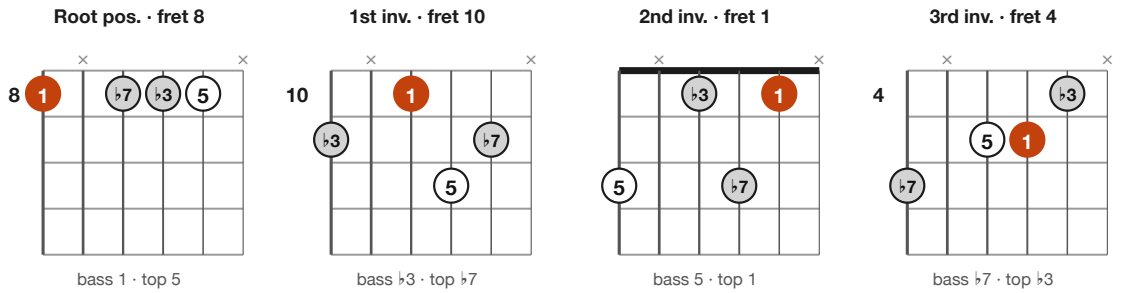
C7
strings 6-4-3-2



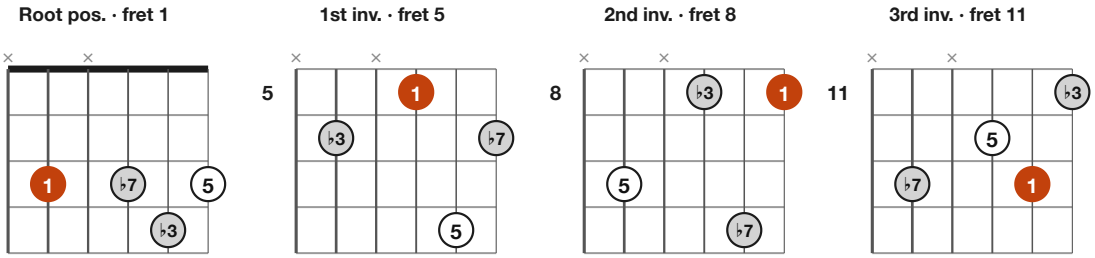
C7
strings 5-3-2-1



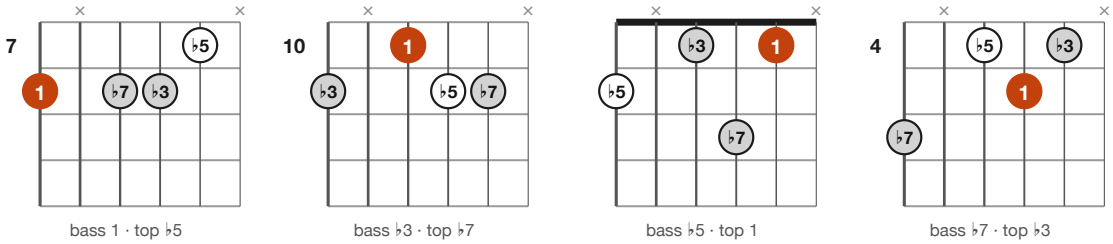
Cm7
strings 6-4-3-2



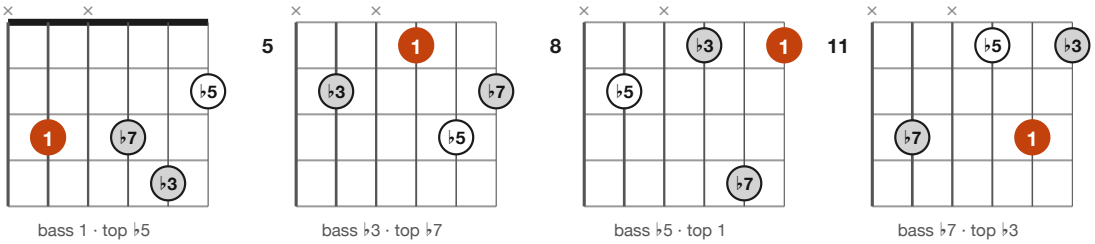
Cm7
strings 5-3-2-1



Cm7♭5
strings 6-4-3-2



Cm7♭5
strings 5-3-2-1

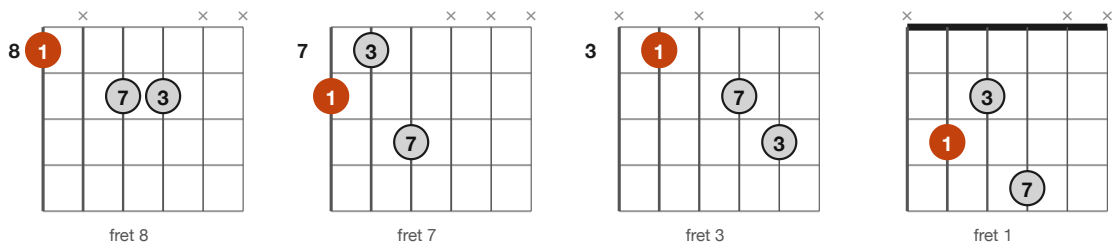


Drop 3 puts a low bass under three close upper voices: a big-band / chord-melody sound. Root-position drop 3 on string 6 and drop 2 root position on string 5 are the two grips most players use to comp every chord in a tune.

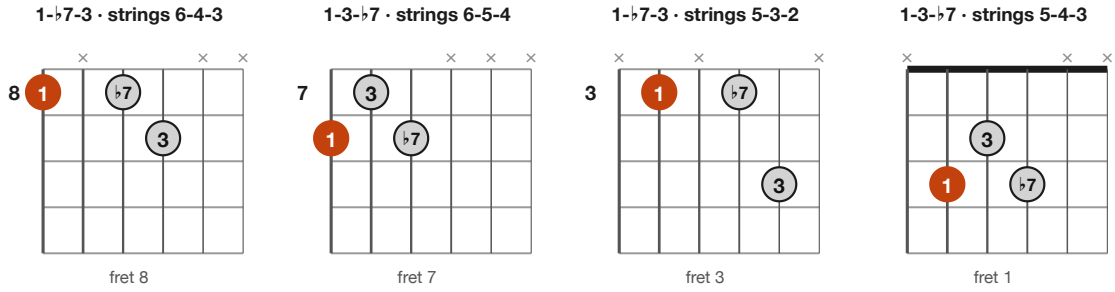
14 Shells, guide tones and extended grips

A **shell** is root + 3rd + 7th (no 5th). The 3rd and 7th — the **guide tones** — define the chord; everything else is colour. Two shapes per root string: **1-7-3** and **1-3-7**.

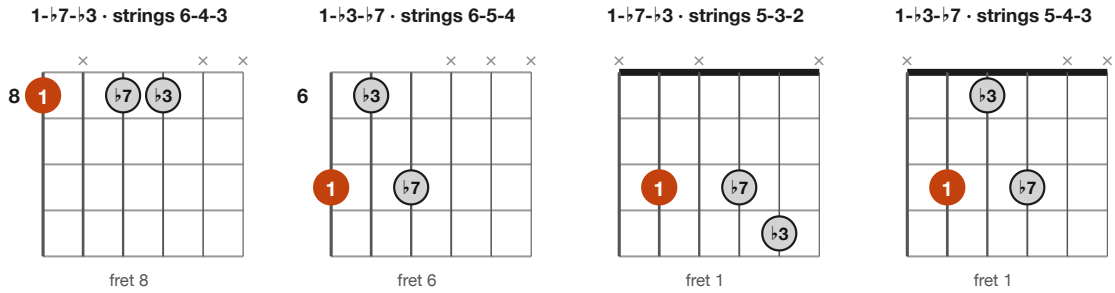
Cmaj7



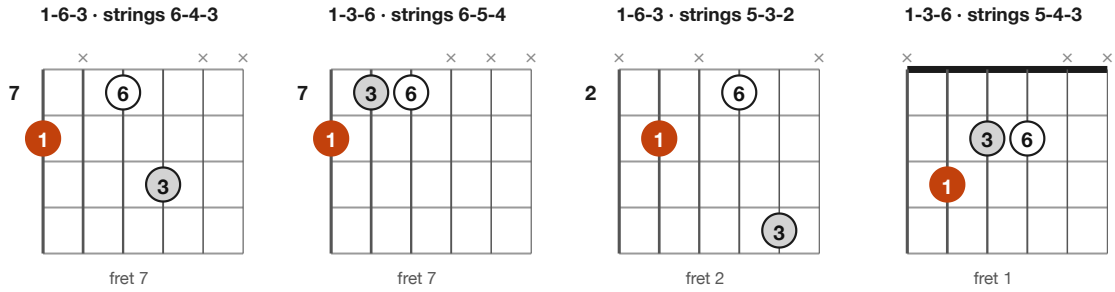
C7



Cm7



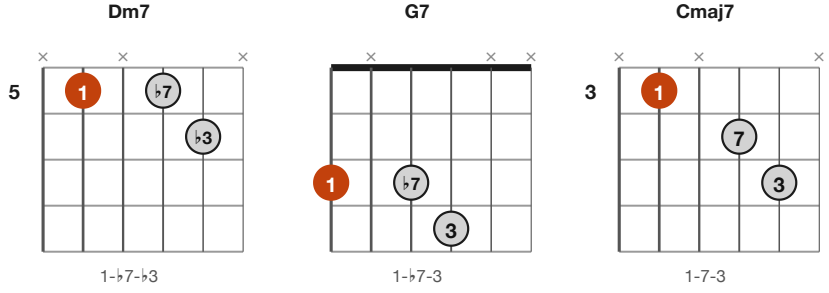
C6
6 replaces the 7



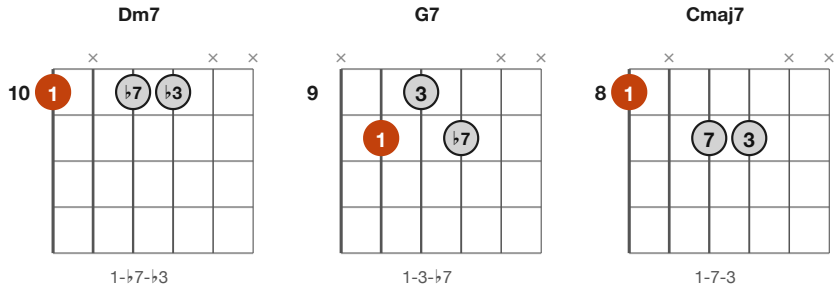
Guide tones through ii–V–I

In a ii–V–I the **7th of each chord falls a half step to the 3rd of the next**, while the 3rd stays put and becomes the next 7th: Dm7 (F, C) → G7 (F, B) → Cmaj7 (E, B). Play only those two notes and you already hear the progression. Alternating root strings (5 → 6 → 5 or 6 → 5 → 6) keeps the hand in one place.

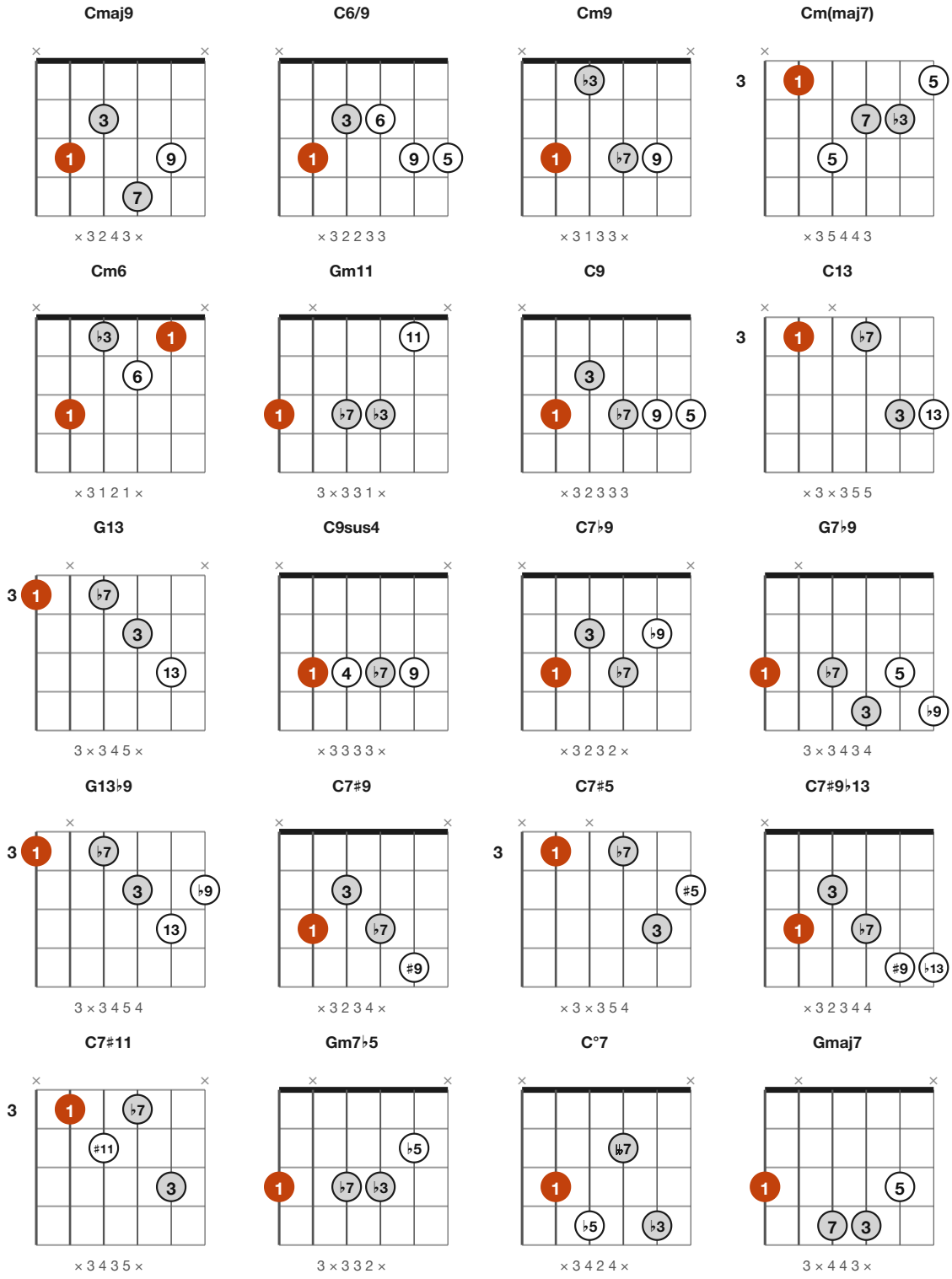
ii–V–I in C
root on 5 → 6 → 5



ii–V–I in C
root on 6 → 5 → 6



Extended and altered grips (labelled by interval — move them anywhere)

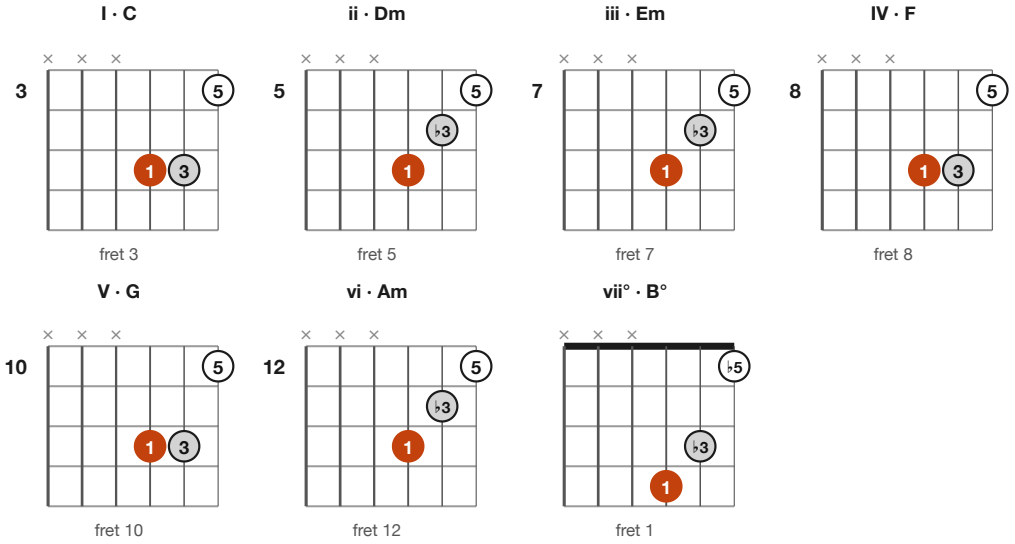


Sub-captions give the frets low E → high e (× = not played). Many of these are **rootless in disguise**: e.g. C9 contains E G B♭ D = Em7♭5 over a C bass; C7♯9 = the "blues 7♯9" with the ♯9 on top. Learn each grip as a set of intervals, then you can alter one note at a time (9 → ♭9 → ♯9, 13 → ♭13).

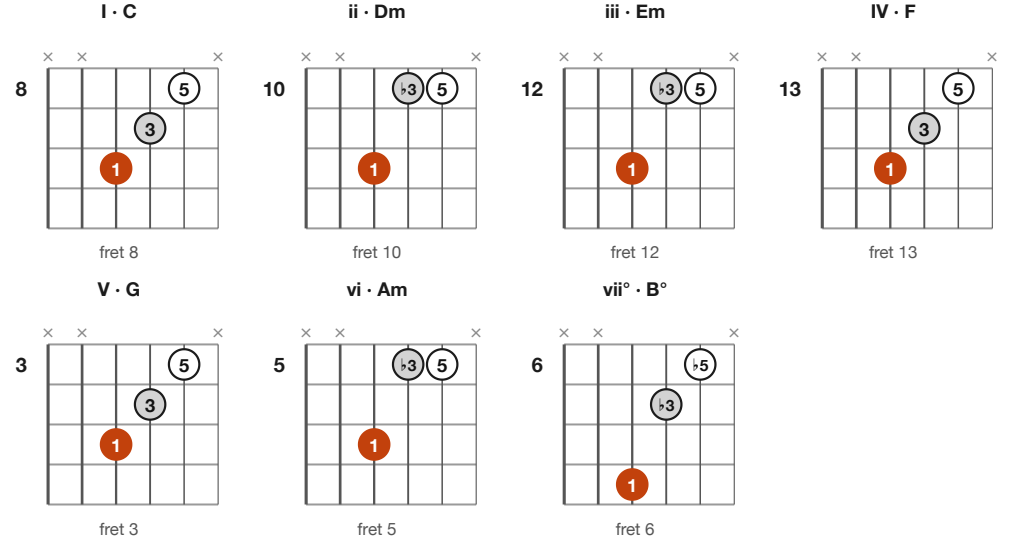
15 The harmonised scale on the neck

The seven chords of C major, each in the same inversion, walking up one string set. This is the bridge between the scale shapes and the chord shapes: every voicing is the previous one with each voice moved up to the next scale note. Say the numeral as you play each chord, then do the same in the other inversions and in other keys.

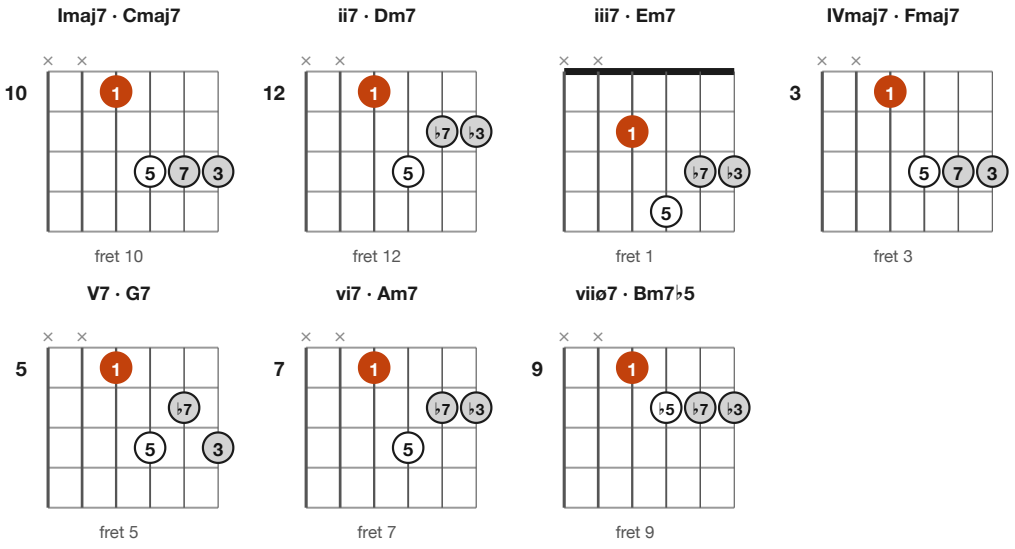
Diatonic triads
root position · strings 3-2-1



Diatonic triads
root position · strings 4-3-2

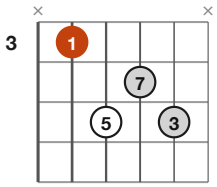


Diatonic 7ths
drop 2, root pos. · strings 4-3-2-1



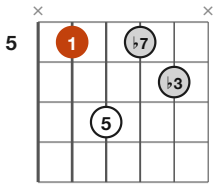
Diatonic 7ths
drop 2, root pos. · strings 5-4-3-2

Imaj7 · Cmaj7



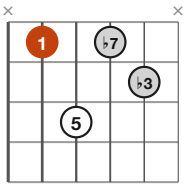
fret 3

ii7 · Dm7



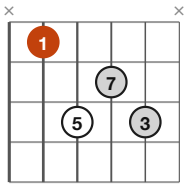
fret 5

iii7 · Em7



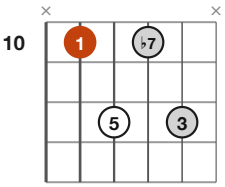
fret 7

IVmaj7 · Fmaj7



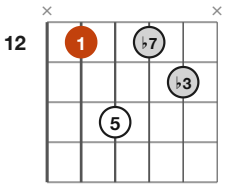
fret 8

V7 · G7



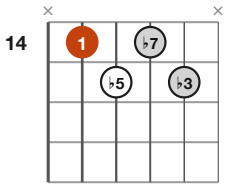
fret 10

vi7 · Am7



fret 12

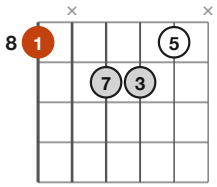
viiø7 · Bm7♭5



fret 14

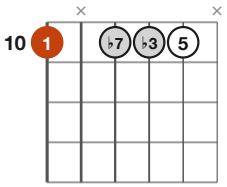
Diatonic 7ths
drop 3, root pos. · strings 6-4-3-2

Imaj7 · Cmaj7



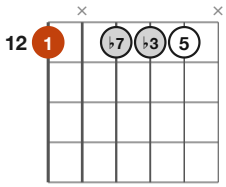
fret 8

ii7 · Dm7



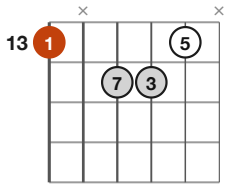
fret 10

iii7 · Em7



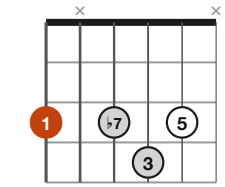
fret 12

IVmaj7 · Fmaj7



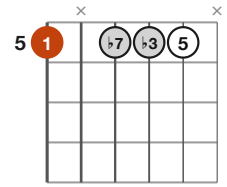
fret 13

V7 · G7



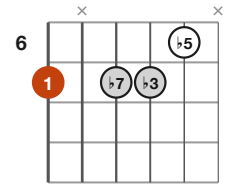
fret 1

vi7 · Am7



fret 5

viiø7 · Bm7♭5



fret 6

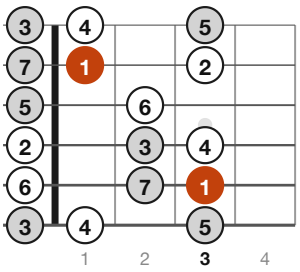
Where a voicing would run off the top of the neck the engine drops it an octave, so a row may wrap back down. Next steps: play the same chord scale in 1st, 2nd and 3rd inversion; play it in the parallel modes (section 9) to hear the borrowed chords.

16 The major scale: CAGED and 3-notes-per-string

Root C. Practise every shape ascending and descending from the lowest note to the highest and back, then from root to root, saying the interval names. Then sequence it (table below).

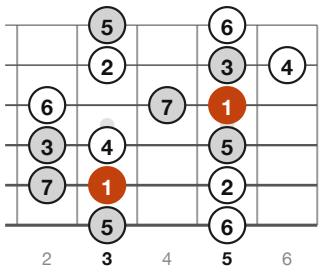
Five CAGED positions (C major)

C shape



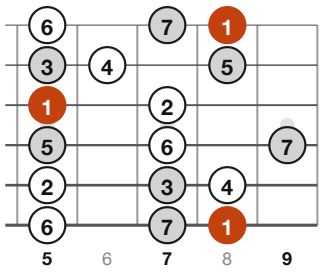
frets 0–4

A shape



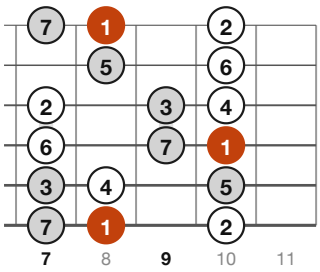
frets 2–6

G shape



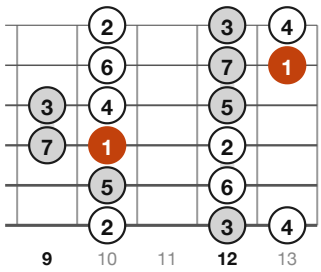
frets 5–9

E shape



frets 7–11

D shape

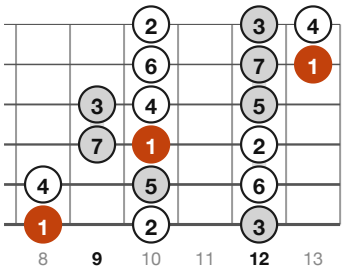


frets 9–13

Grey = the Cmaj7 arpeggio (3 5 7) inside the scale. Each position overlaps the next; together they cover the neck (see section 10). The same five shapes, with a different note called "1", are the five positions of every mode of the major scale.

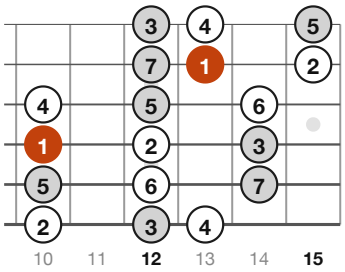
Seven three-notes-per-string patterns (C major)

Pattern 1: starts on the 1 (string 6)



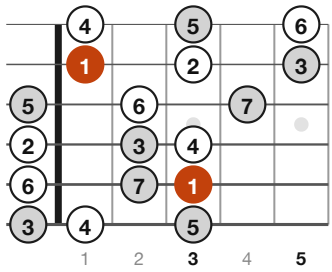
frets 8–13

Pattern 2: starts on the 2 (string 6)



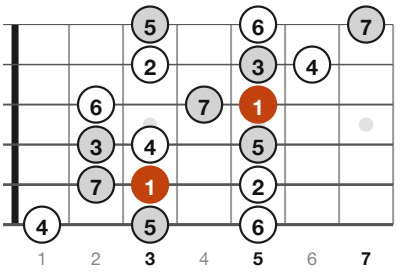
frets 10–15

Pattern 3: starts on the 3 (string 6)



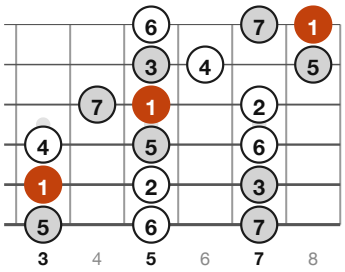
frets 0–5

Pattern 4: starts on the 4 (string 6)



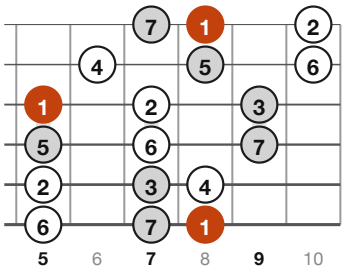
frets 1–7

Pattern 5: starts on the 5 (string 6)



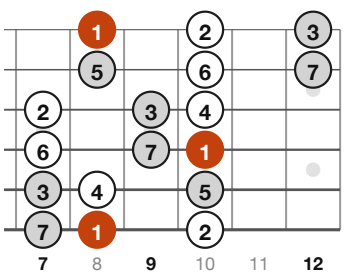
frets 3–8

Pattern 6: starts on the 6 (string 6)



frets 5–10

Pattern 7: starts on the 7 (string 6)



frets 7–12

Three notes on every string: even picking groups, wide reach, easy legato. Pattern n starts on scale degree n on string 6; each pattern is also the "home" shape of a mode (pattern 2 = D Dorian, pattern 5 = G Mixolydian…). Each pattern is shown in its lowest position; it repeats 12 frets higher.

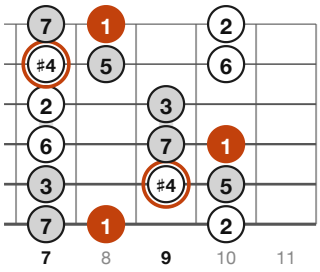
Sequences — play every shape through these

Sequence	In C (ascending; then descend the same way)
3rds	C E · D F · E G · F A · G B · A C · B D …
4ths	C F · D G · E A · F B · G C · A D · B E …
Groups of 3	C D E · D E F · E F G · F G A · G A B …
Groups of 4	C D E F · D E F G · E F G A · F G A B · G A B C …
Triads (diatonic arpeggios)	C E G · D F A · E G B · F A C · G B D …
7th chords (diatonic arpeggios)	C E G B · D F A C · E G B D · F A C E …
1-2-3-5 cells	C D E G · D E F A · E F G B · F G A C …

17 Modes on one root: the parallel view

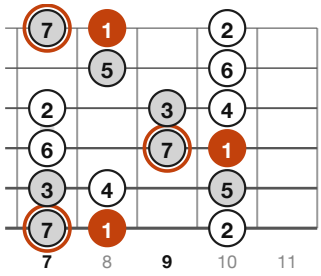
All seven modes of the major scale on the **same root** (C, root on string 6 at fret 8), brightest to darkest. Ringed = the note that gives the mode its colour. Play them back to back over a C drone and listen to one note change at a time. This is the "parallel" view you need for modal interchange; the CAGED/3NPS shapes on the previous page are the "derivative" view (one parent scale, many starting notes).

C Lydian



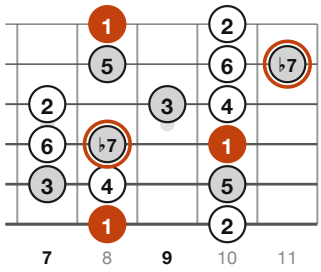
1 2 3 #4 5 6 7 · colour: #4

C Ionian (major)



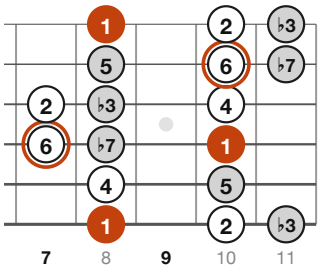
1 2 3 4 5 6 7 · colour: 7

C Mixolydian



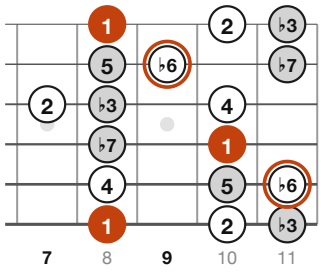
1 2 3 4 5 6 ♭7 · colour: ♭7

C Dorian



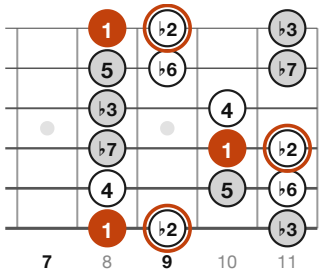
1 2 ♭3 4 5 6 ♭7 · colour: 6

C Aeolian (natural minor)



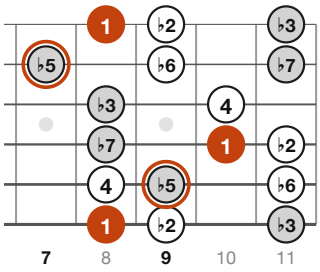
1 2 ♭3 4 5 ♭6 ♭7 · colour: ♭6

C Phrygian



1 ♭2 ♭3 4 5 ♭6 ♭7 · colour: ♭2

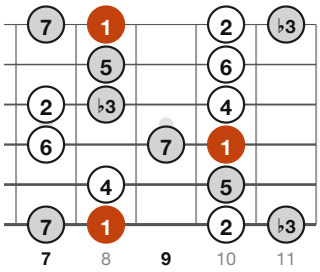
C Locrian



1 ♭2 ♭3 4 ♭5 ♭6 ♭7 · colour: ♭5

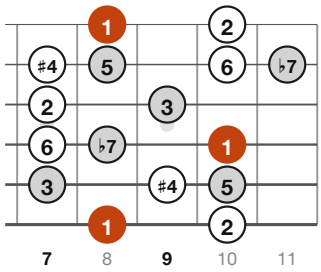
The most-used melodic and harmonic minor modes, same position

C Melodic minor



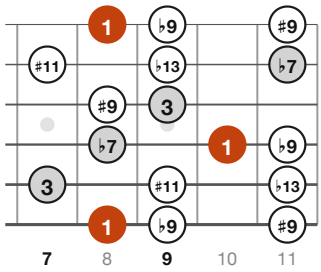
1 2 ♭3 4 5 6 7

C Lydian dominant



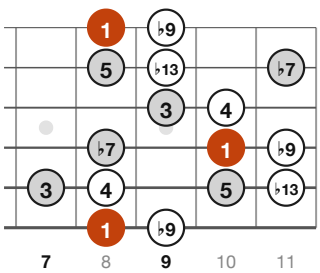
1 2 3 #4 5 6 ♭7

C Altered



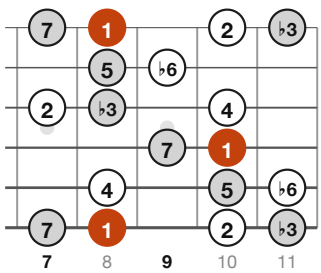
1 ♭9 #9 3 #11 ♭13 ♭7

C Phrygian dominant



1 ♭9 3 4 5 ♭13 ♭7

C Harmonic minor



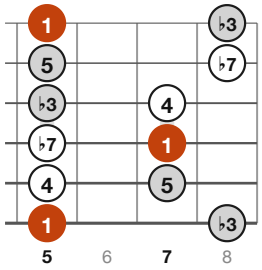
1 2 ♭3 4 5 ♭6 7

18 Pentatonic and blues boxes

Grey = the triad tones (minor ♭3 and 5, major 3 and 5). The pentatonic is a major/minor scale with the two "difficult" notes removed: no half steps, nothing clashes.

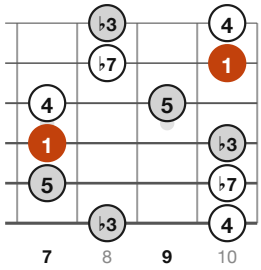
Minor pentatonic (A): 1 ♭3 4 5 ♭7

Box 1



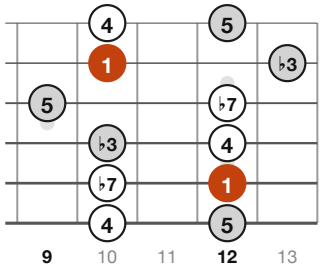
frets 5–8

Box 2



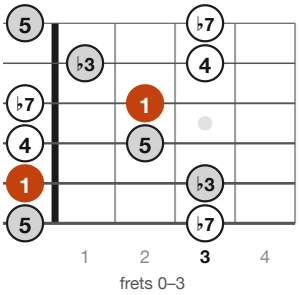
frets 7–10

Box 3



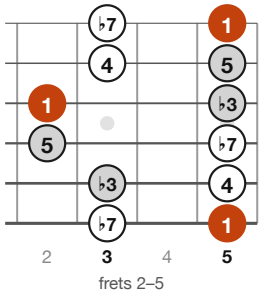
frets 9–13

Box 4



frets 0–3

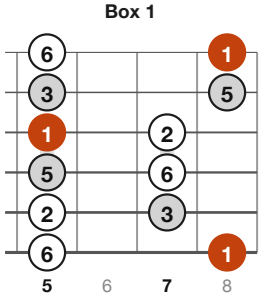
Box 5



frets 2–5

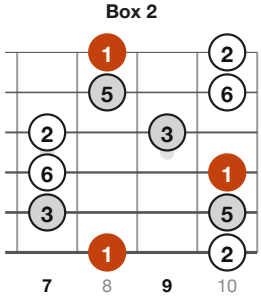
Major pentatonic (C): 1 2 3 5 6 — the same five shapes, root moved to the relative major

Box 1



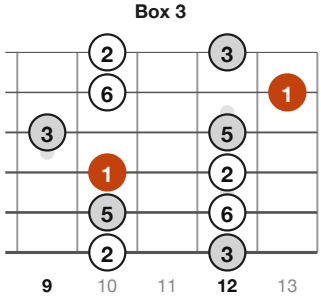
frets 5–8

Box 2



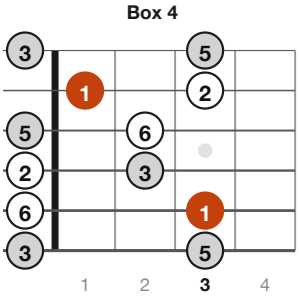
frets 7–10

Box 3



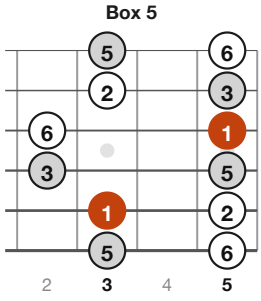
frets 9–13

Box 4



frets 0–3

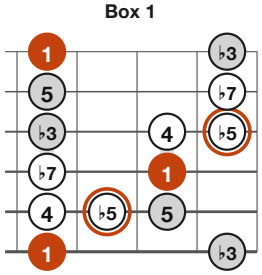
Box 5



frets 2–5

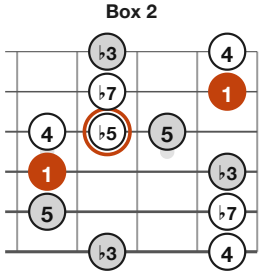
Blues scale (A): minor pentatonic + ♭5 (ringed)

Box 1



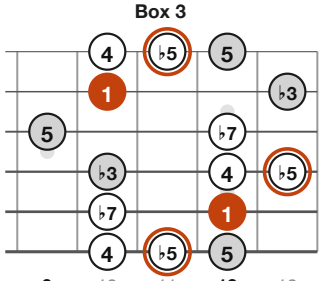
frets 5–8

Box 2

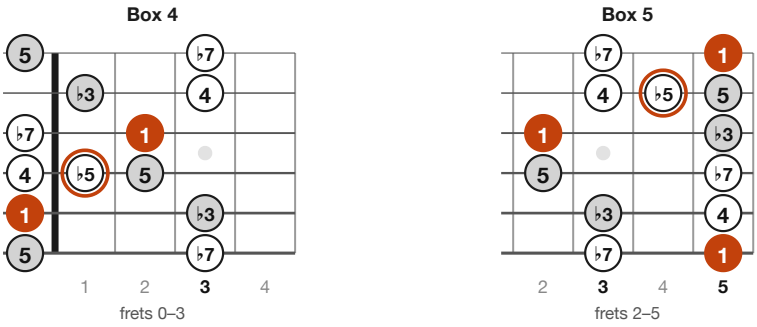


frets 7–10

Box 3



frets 9–13

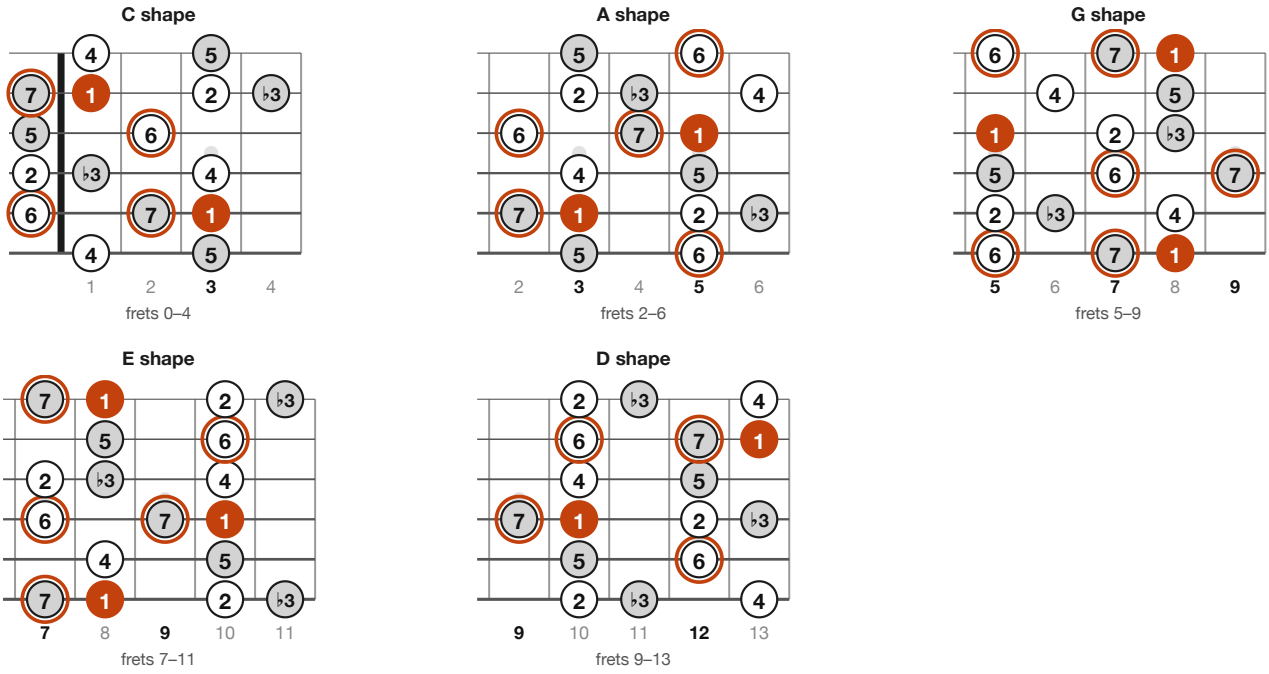


Box 1 is the familiar root-on-string-6 shape. Boxes connect end to end (box 1's top notes are box 2's bottom notes) and repeat after box 5. The major-pentatonic boxes are the minor ones with the root moved: A minor pentatonic = C major pentatonic. Over a dominant blues, mix the two (♭3 → 3 slides) and add the ♭5.

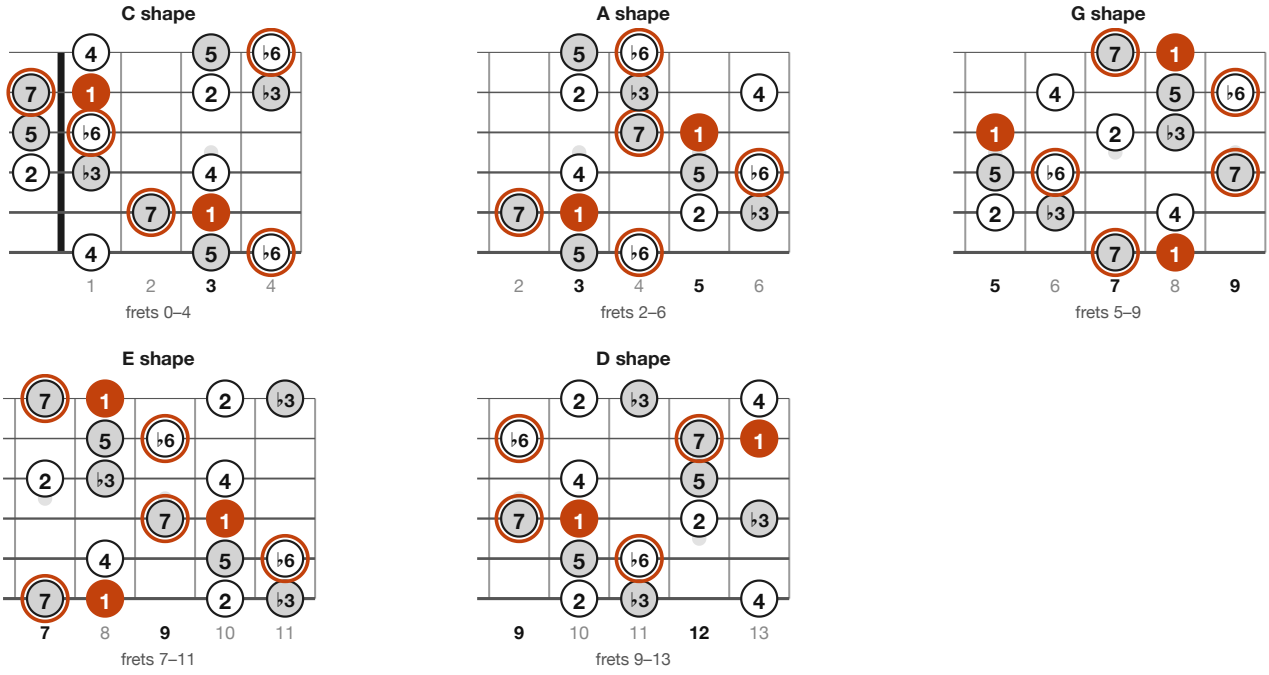
19 Melodic minor and harmonic minor

Grey = the tonic Cm(maj7) arpeggio (♭3 5 7) inside each scale.

C melodic minor: 1 2 ♭3 4 5 6 7 (ringed: the ♮6 and ♮7 that differ from natural minor)



C harmonic minor: 1 2 ♭3 4 5 ♭6 7 (ringed: the ♭6–7 augmented 2nd)

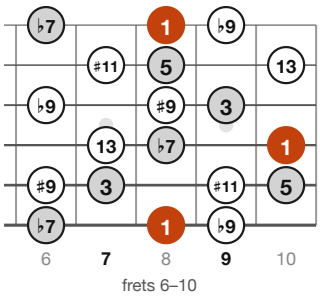


Same five CAGED windows as the major scale. Use them through their modes: **G altered = A♭ melodic minor**, **F Lydian dominant = C melodic minor**, **A Locrian ♮2 = C melodic minor**, **G Phrygian dominant = C harmonic minor**. Know the parent shape, then think of the chord's root inside it.

20 Symmetric scales: diminished and whole tone

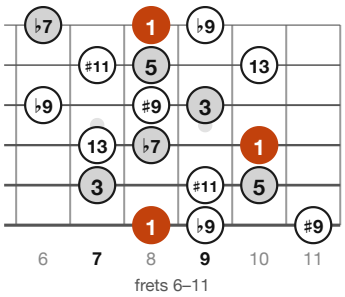
Half-whole diminished (C): use over C7♭9 / C13♭9

One position



frets 6–10

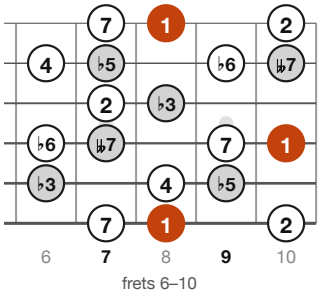
Three notes per string



frets 6–11

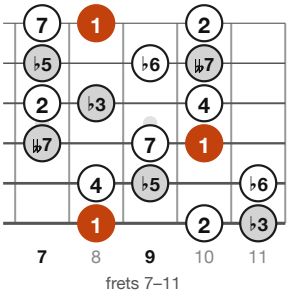
Whole-half diminished (C): use over C°7

One position



frets 6–10

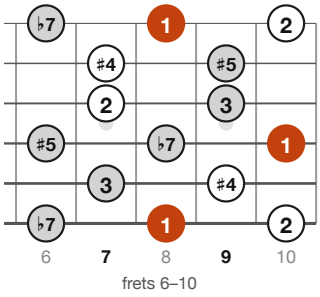
Three notes per string



frets 7–11

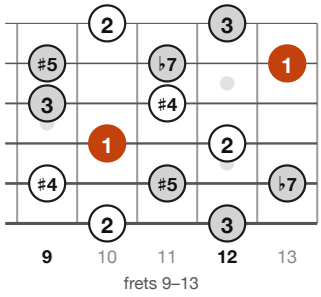
Whole tone (C): use over C7♯5 / C9♯5

One position



frets 6–10

Next position (two frets up)



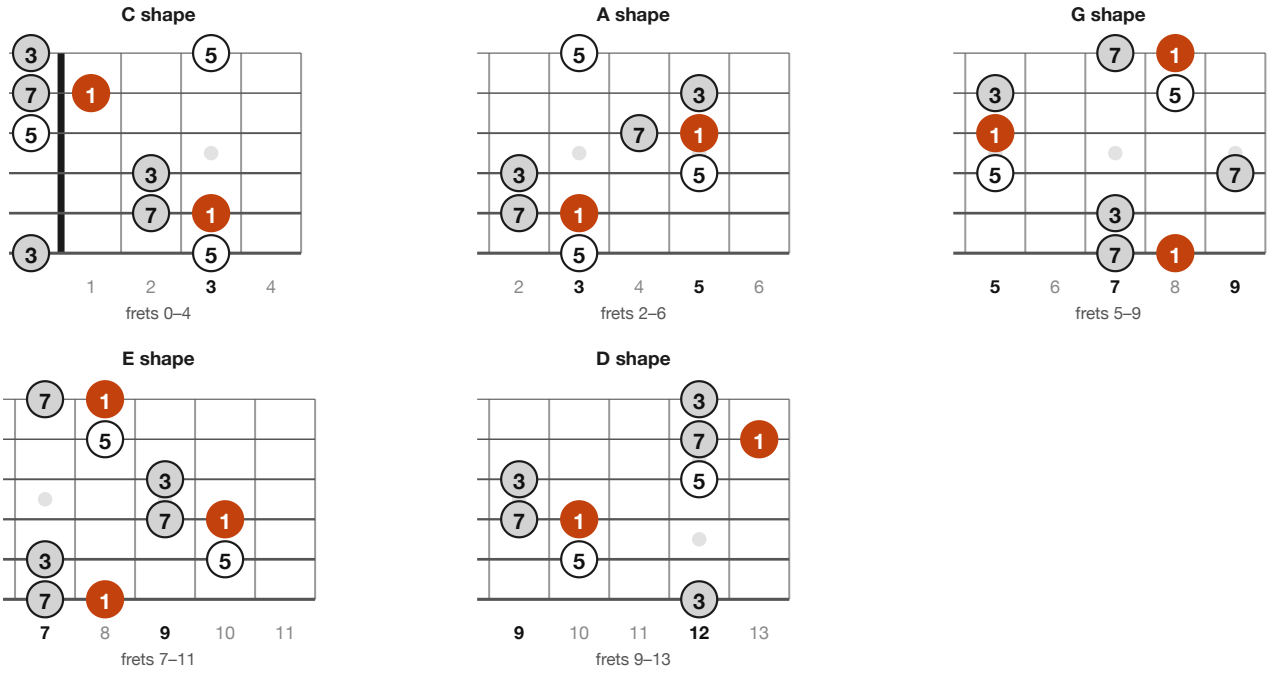
frets 9–13

Half-whole is labelled as tensions over C7 (♭9 ♯9 ♯11 13). Grey = the chord the scale is used over (C7 for half-whole, C°7 for whole-half, C7♯5 for whole tone). **Diminished: move any shape 3 frets and it is the same scale** (the root changes to another chord tone). **Whole tone: move any shape 2 frets.** So there are only three diminished scales and two whole-tone scales to learn.

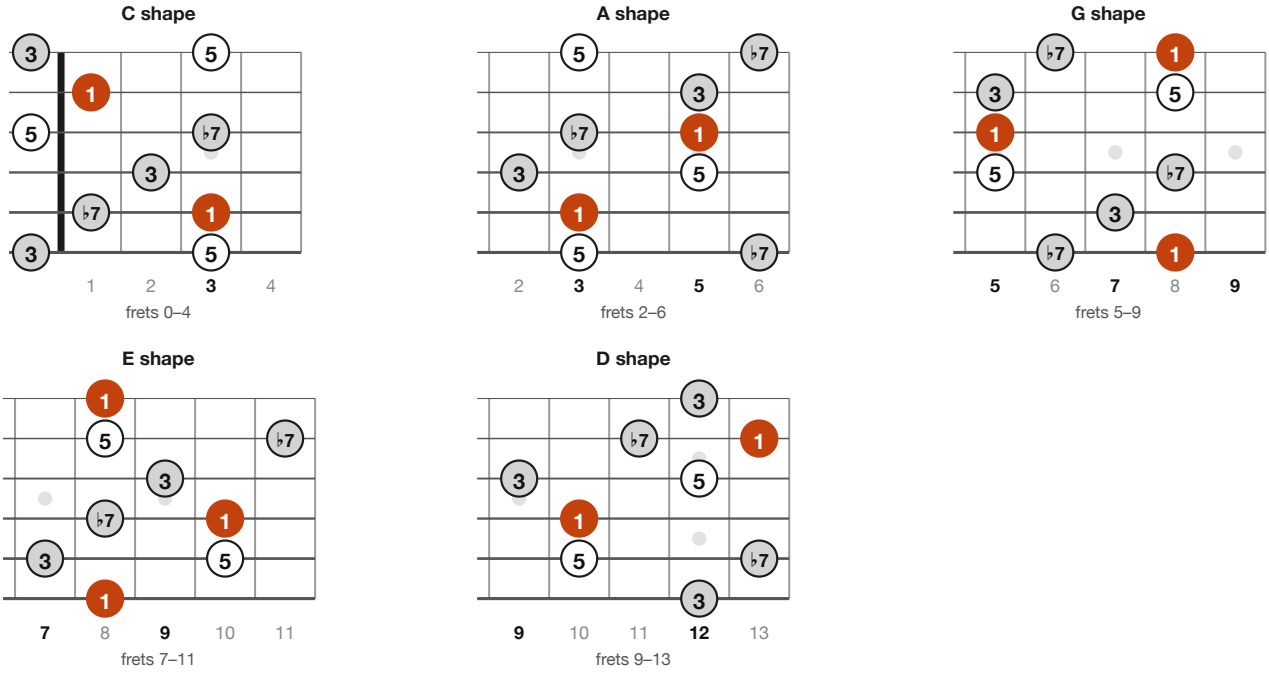
21 Seventh-chord arpeggios in the five CAGED shapes

Root C, grey = 3rd and 7th. The arpeggio is the skeleton of every scale shape in the same window: these are the notes to land on.

Cmaj7 arpeggio · 1 3 5 7 (C E G B)

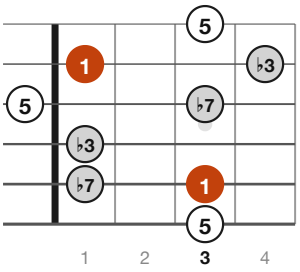


C7 arpeggio · 1 3 5 ♭7 (C E G B♭)



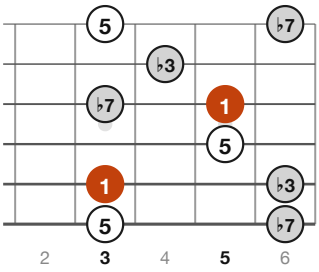
Cm7 arpeggio · 1 ♭3 5 ♭7 (C E♭ G B♭)

C shape



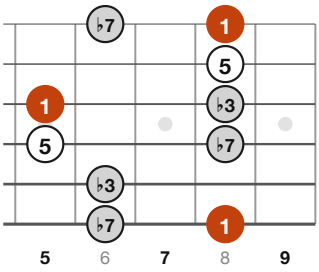
frets 0–4

A shape



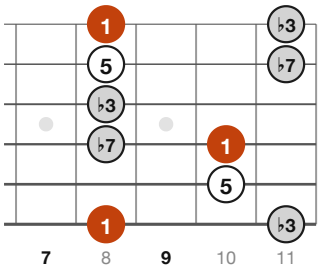
frets 2–6

G shape



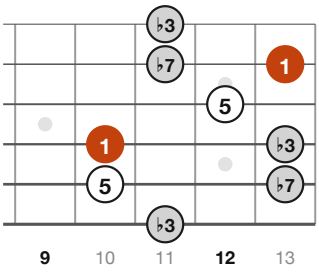
frets 5–9

E shape



frets 7–11

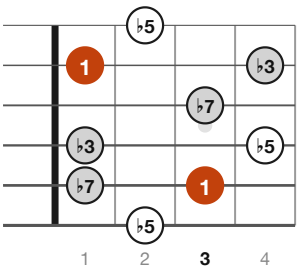
D shape



frets 9–13

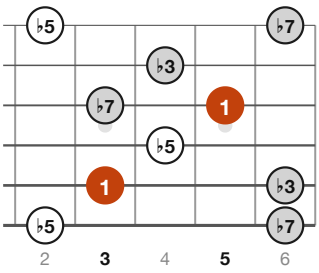
Cm7♭5 arpeggio · 1 ♭3 ♭5 ♭7 (C E♭ G♭ B♭)

C shape



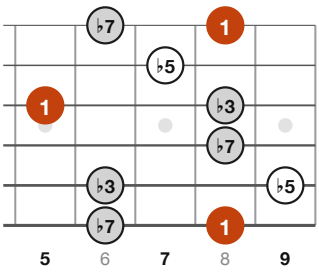
frets 0–4

A shape



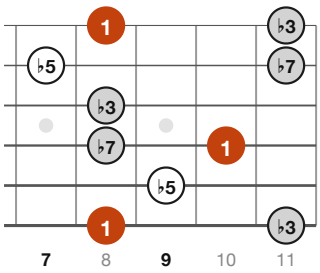
frets 2–6

G shape



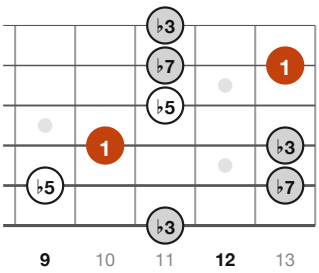
frets 5–9

E shape



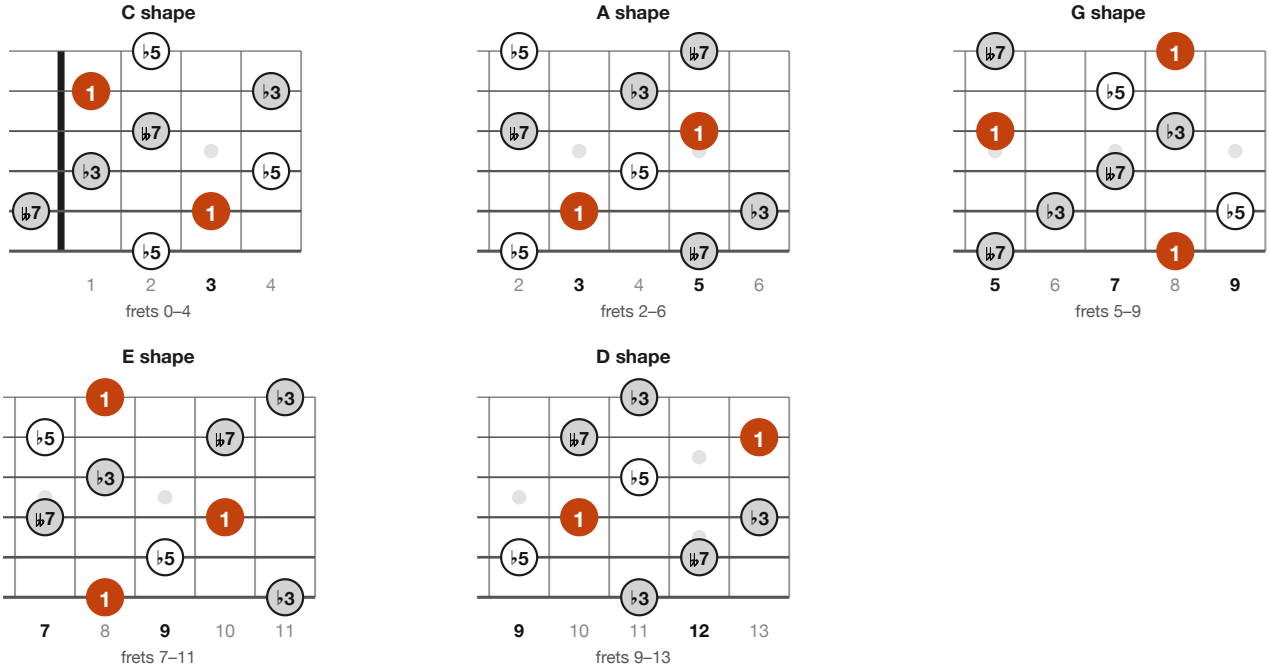
frets 7–11

D shape



frets 9–13

C°7 arpeggio · 1 ♭3 ♭5 𝄫7 (C E♭ G♭ B𝄫)



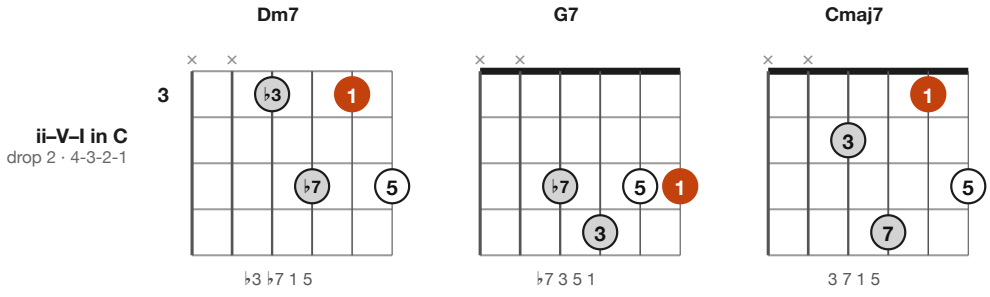
Practise each arpeggio from the lowest chord tone in the window to the highest and back, then from the root, then starting on the 3rd, 5th and 7th. Then cycle the qualities in one shape (maj7 → 7 → m7 → m7♭5 → °7): each step lowers one note by a fret.

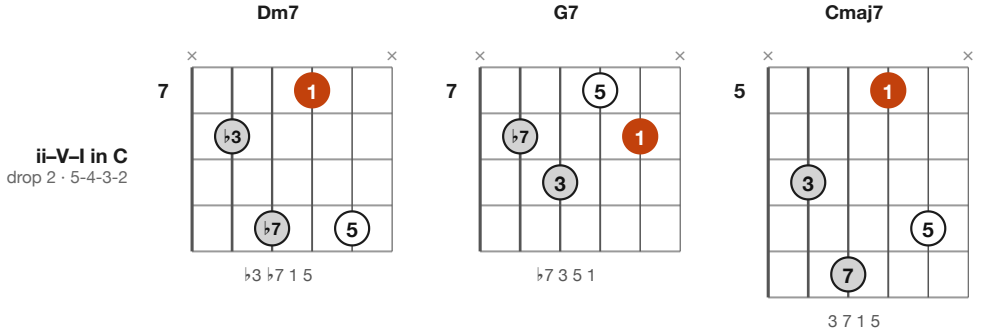
22 Voice leading and chord melody

Principles

- **Keep common tones** in the same voice; move every other voice by the **smallest step** available (half or whole step).
- **Guide tones (3rd and 7th) carry the harmony.** When the root moves up a 4th (ii–V, V–I, any cycle of 4ths) the 7th falls a step to the next 3rd and the 3rd holds to become the next 7th (between two dominant 7ths it slips down a half step instead).
- **Contrary motion** between bass and top voice sounds strongest; parallel motion in all voices sounds like sliding shapes (fine as an effect, weak as voice leading).
- In any one area of the neck, choose the **inversion** of the next chord that is closest to the current one — never jump to the "root-position" grip by habit.

ii–V–I voice-led (engine-chosen nearest inversions)





Guide-tone line through the cycle of 4ths (dominant 7ths)

	1	2	3	4	5	6	7	8	9	10	11	12
Chord	**C7**	**F7**	**B♭7**	**E♭7**	**A♭7**	**D♭7**	**G♭7**	**B7**	**E7**	**A7**	**D7**	**G7**
Upper voice	B♭ (♭7)	A (3)	A♭ (♭7)	G (3)	G♭ (♭7)	F (3)	F♭ (♭7)	D♯ (3)	D (♭7)	C♯ (3)	C (♭7)	B (3)
Lower voice	E (3)	E♭ (♭7)	D (3)	D♭ (♭7)	C (3)	C♭ (♭7)	B♭ (3)	A (♭7)	G♯ (3)	G (♭7)	F♯ (3)	F (♭7)

With dominant 7ths all the way round, **both voices slide down a half step at every change**: the ♭7 becomes the next 3rd and the 3rd becomes the next ♭7, so the two lines swap roles each bar. Play them on strings 4-3 or 3-2 through all 12 dominants without moving more than one fret.

Chord melody: the melody note goes on top

Melody note on top	1	3	5	7
Drop 2, strings 4-3-2-1	3rd inv. (7 in bass)	Root pos. (1 in bass)	1st inv. (3 in bass)	2nd inv. (5 in bass)
Drop 3, strings 5-3-2-1	2nd inv. (5 in bass)	3rd inv. (7 in bass)	Root pos. (1 in bass)	1st inv. (3 in bass)

- **1. Learn the melody** on the top two strings in one position; sing it.
- **2. Strong beats get a chord** with the melody note as the highest voice: look up which inversion puts that chord tone on top (table above). Drop 3 adds a bass note under the melody; drop 2 keeps it compact.
- **3. Melody note not a chord tone?** Treat it as a tension: 9 replaces the root, 13 replaces the 5th, 11 / ♯11 replaces the 3rd on minor / major chords. Or harmonise it with a diatonic or diminished **passing chord**.
- **4. Between strong beats** play single melody notes, or move inner voices. Bass notes on beat 1 and 3 imply the harmony with very little effort.
- **5. Reharmonise** (next section) only after the plain version sounds smooth.

23 Reharmonisation essentials

Technique	Rule	Before	After (in C)
Diatonic substitution	Chords sharing two notes can replace each other: I ↔ iii ↔ vi, IV ↔ ii.	Imaj7 | IVmaj7 | ii7 | V7 Cmaj7 | Fmaj7 | Dm7 | G7	**iii7 | ii7 | ii7 | V7** Em7 | Dm7 | Dm7 | G7
Secondary dominant	Precede any chord with its own V7 (a 5th above).	Imaj7 | ii7 | V7 | Imaj7 Cmaj7 | Dm7 | G7 | Cmaj7	**Imaj7 VI7 | ii7 | V7 | Imaj7** Cmaj7 A7 | Dm7 | G7 | Cmaj7
Related ii	Split any dominant into its ii–V.	Imaj7 | VI7 | ii7 V7 | Imaj7 Cmaj7 | A7 | Dm7 G7 | Cmaj7	**Imaj7 | iii7 VI7 | ii7 V7 | Imaj7** Cmaj7 | Em7 A7 | Dm7 G7 | Cmaj7
Tritone substitution	Replace V7 with the dominant a ♭5 away: bass falls by half steps.	ii7 | V7 | Imaj7 Dm7 | G7 | Cmaj7	**ii7 | ♭II7 | Imaj7** Dm7 | D♭7 | Cmaj7
Backdoor dominant	Approach I from ♭VII7 (with its ii: iv7).	ii7 | V7 | Imaj7 Dm7 | G7 | Cmaj7	**iv7 | ♭VII7 | Imaj7** Fm7 | B♭7 | Cmaj7
Passing diminished	A dim7 a half step below the target fills a whole-step bass gap.	Imaj7 | ii7 | iii7 Cmaj7 | Dm7 | Em7	**Imaj7 #i°7 | ii7 #ii°7 | iii7** Cmaj7 C#°7 | Dm7 D#°7 | Em7
Modal interchange	Borrow from the parallel minor (section 9).	Imaj7 | IVmaj7 | Imaj7 Cmaj7 | Fmaj7 | Cmaj7	**Imaj7 | ivm6 | Imaj7** Cmaj7 | Fm6 | Cmaj7
Side-step (chromatic approach chord)	Approach any chord from a half step above or below with the same quality.	ii7 | V7 | Imaj7 Dm7 | G7 | Cmaj7	**♭iii7 ii7 | ♭II7 V7 | Imaj7** E♭m7 Dm7 | D♭7 G7 | Cmaj7
Line cliché	Hold the chord; move one inner voice chromatically.	i | % | % | % Cm | Cm | Cm | Cm	**i | i(maj7) | i7 | im6** Cm | Cm(maj7) | Cm7 | Cm6
Pedal point	Keep the bass on one note (usually 1 or 5) while the chords above change.	Imaj7 | IVmaj7 | V7 | Imaj7 Cmaj7 | Fmaj7 | G7 | Cmaj7	**Imaj7 | IVmaj7/5 | V7sus4 | Imaj7** Cmaj7 | Fmaj7/C | G7sus4 | Cmaj7
Extend and alter	Same function, richer chord: maj7 → maj9/6/9, m7 → m9/m11, V7 → 13 / 7♭9 / 7alt.	ii7 | V7 | Imaj7 Dm7 | G7 | Cmaj7	**ii9 | V7alt | I6/9** Dm9 | G7alt | C6/9

Order of operations

- **Keep the melody.** A reharmonisation must still contain (or tension-colour) the melody note on each strong beat. Check: is the melody a chord tone or a usable tension of the new chord?
- **Keep the arrival points** (cadences, the tonic at phrase ends); reharmonise the road, not the destination.
- **Change one thing at a time**, play it, and decide by ear. Two or three substitutions per chorus is plenty.
- Every substitution is a **voice-leading** device: the good ones move the bass or a guide tone by a half step.

24 Improvisation essentials

The note hierarchy

Level	Notes	Where they go
Chord tones	1 3 5 7 (3rd and 7th most of all)	on the beat, at the start and end of phrases, on chord changes
Tensions	9 11 13 and alterations, from the chord–scale	held for colour, or resolved to a neighbouring chord tone
Passing / approach notes	any note a half or whole step from a target	off the beat, moving into a chord tone

Target tones, approach notes and enclosures (targets: E = 3rd of C, G = 5th of C)

Device	Into E	Into G
Chromatic approach from below	D♯ → E	F♯ → G
Scale approach from above	F → E	A → G
Enclosure: above, then below	F → D♯ → E	A → F♯ → G
Enclosure: below, then above	D♯ → F → E	F♯ → A → G
Double chromatic from below	D → D♯ → E	F → F♯ → G
Double chromatic from above	F♯ → F → E	A → A♭ → G
Scale above + chromatic below	F → D → D♯ → E	A → F → F♯ → G

Aim each phrase at a **target** (usually the 3rd or 7th of the next chord) and arrive **on the beat**: start the approach figure early enough that the target lands on beat 1 (or 3). The **bebop scales** add one chromatic passing note so that a run of 8th notes starting on a chord tone on the beat keeps chord tones on every beat.

Arpeggio superimposition: one shape, richer sound

Over	Play the arpeggio of	Which gives	Sound
Cmaj7	**Em7** (E G B D)	3 5 7 9	Cmaj9 without the root
Cmaj7	**Gmaj7** (G B D F♯)	5 7 9 ♯11	Lydian brightness
Cmaj7	**Bm7** (B D F♯ A)	7 9 ♯11 13	maj13♯11, very open
Dm7	**Fmaj7** (F A C E)	♭3 5 ♭7 9	Dm9
Dm7	**Am7** (A C E G)	5 ♭7 9 11	Dm11, open
G7	**Bm7♭5** (B D F A)	3 5 ♭7 9	G9
G7	**Dm7** (D F A C)	5 ♭7 9 11	G9sus / 11 colour
G7	**B°7** (B D F A♭)	3 5 ♭7 ♭9	G7♭9
G7	**D♭7** (D♭ F A♭ C♭)	♯11 ♭7 ♭9 3	tritone sub: ♯11 and ♭9
G7alt	**B♭m7** (B♭ D♭ F A♭)	♯9 ♯11 ♭7 ♭9	altered
G7alt	**A♭m(maj7)** (A♭ C♭ E♭ G)	♭9 3 ♭13 1	altered (melodic minor ½ step up)
Bm7♭5	**Dm7** (D F A C)	♭3 ♭5 ♭7 ♭9	Locrian, adds ♭9
Cm(maj7)	**E♭maj7♯5** (E♭ G B D)	♭3 5 7 9	melodic-minor colour with 9

Rule of thumb: an arpeggio built on the **3rd** of a chord gives 3 5 7 9; on the **5th** gives 5 7 9 11/♯11; on the **7th** gives 7 9 11 13. Over a dominant, the **dim7 on the 3rd** gives 7♭9 and the **tritone sub's dominant** gives ♯11 ♭7 ♭9 3.

Motif development

- **Repeat** it — then change one thing.
- **Sequence:** move it up/down the scale or by a fixed interval.
- **Displace** it rhythmically: start it on beat 2, or an 8th later.
- **Augment / diminish:** double or halve the note values.
- **Invert** the contour (up becomes down); **retrograde** (backwards).
- **Fragment:** keep only its first two or three notes and build from them.
- **Call and response:** a two-bar phrase, then a two-bar answer.

Playing over changes — build it in layers

- **1. Roots** in time (one per chord), then **guide tones** (3rds and 7ths) as a slow melody.
- **2. Arpeggios**, nearest inversion to the previous one, connecting at the change.
- **3. Arpeggio + approach notes** into each next chord tone.
- **4. Scales** linking chord tones; land a chord tone on each change.
- **5. Superimpositions, tensions, side-slipping** and outside-inside lines.
- **6. Motifs** carried across the changes: same idea, adjusted to each chord.

Phrasing

- Leave space: phrases of 2–4 bars with rests between. Rests are where the listener digests the idea.
- Vary where phrases start (beat 1, the "and" of 1, beat 2, the "and" of 4 of the previous bar) and how long they are.
- Sing the phrase, then play it. If you cannot sing it, simplify it.
- Dynamics, articulation (slides, hammer-ons, ghost notes), vibrato and note length are half of the message.

25 Rhythm and time

Subdivisions

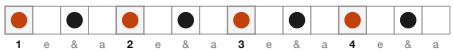
Subdivision	Count it	Per beat
Quarters	1 2 3 4	1
8ths	1 & 2 &	2
8th triplets	1 trip let 2 trip let	3
16ths	1 e & a 2 e & a	4
Quintuplets	1 ta ka di mi	5
16th triplets (sextuplets)	1 la li & la li	6

Feels

Feel	What it means
Straight	each beat split evenly (1 : 1)
Swing / shuffle	each beat long–short, about 2 : 1 — the "1" and "let" of 1-trip-let; lighter (closer to straight) at fast tempos
Backbeat	accents on 2 and 4
Syncopation	accents off the beat: the "&", "e", "a" — or tied over the beat
Half-time / double-time	the pulse feels half or twice as fast; the chords do not change speed

Patterns to know (16 steps = one 4/4 bar of 16ths; accent colour = accent)

Straight 8th strum, backbeat



down on each beat, up on each "&"; accent 2 and 4

Pop strum



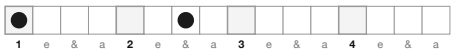
D · D U · U D U (1, 2 &, & 4 &)

Funk 16ths



keep the hand moving in 16ths; hit only some

Charleston



swing comping: beat 1 and the "&" of 2

Reverse Charleston



the "&" of 1 and beat 3

Four to the bar (Freddie Green style)



short quarter notes, 2 and 4 slightly stronger

Son clave 3-2



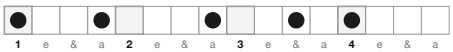
the backbone of Afro-Cuban music

Son clave 2-3



same pattern starting on the other bar

Rumba clave 3-2



third note pushed late

Bossa nova clave



the guitar's thumb plays 1 and 3 underneath

Tresillo (3+3+2), twice



Latin, second-line, reggaeton, pop

Shuffle / swing 8ths



triplet grid: play the 1st and 3rd of each triplet

Time-feel practice

- **Metronome on 2 and 4:** set the click to half the tempo and hear it as beats 2 and 4 (the hi-hat in swing). Count out loud until it locks.
- **Gap click:** click only on beat 1, then only on beat 1 of every second bar, then every fourth. Your internal time fills the gaps.
- **Subdivide every tempo:** play quarters, 8ths, triplets, 16ths against the same click; switch on command without losing the beat.
- **Slow is harder:** practise at 40–60 bpm as well as fast; at slow tempos every note placement shows.
- **Tempo ladder:** start where it is perfect, go up 4–6 bpm per success, back down 10 when it breaks. Stop at 3 clean repetitions.
- **Comping time:** play only the rhythm (muted strings) with the click; then add chords without changing the feel.
- **Record yourself** against the click and listen back: are you ahead, behind, or on it? Then do it on purpose.

26 Ear training

Intervals

- Sing every interval up and down from any note before you play it; use the reference tunes in section 2 only as a bridge.
- Recognise intervals **in a key** (as scale degrees) rather than in isolation: hear 5 → 1, 7 → 1, 4 → 3.
- Do melodic (one after the other) and harmonic (together) intervals; harmonic ones are what you hear inside chords.
- On the guitar: play a root, sing a target interval, then find it on the neck. Check. Repeat in every position.

Chord qualities

Chord	Hear it as
Major	bright, stable
Minor	dark, stable
Diminished	tense, closed
Augmented	floating, unresolved
sus4	suspended, wants the 3rd
maj7	lush; the 7 rubs against the root
7	bluesy, pulls to resolve
m7	soft minor
m7♭5	dark, yearning
°7	dramatic, symmetrical

Function: hearing where you are in the key

- **Sing the bass line.** Most progressions are recognisable from the bass: 1–4–5–1, 1–6–2–5, 1–♭7–4, the chromatic descents.
- **Hear degrees, not letters:** play a cadence in a key, then a single note; name its degree (do–re–mi or 1–2–3). Ten a day, random keys.
- **Chord numerals:** I and vi share two notes (tonic family); ii and IV (predominant) "lean"; V and vii° pull. Name the function before the exact chord.
- **Colour notes:** borrowed iv (the ♭6), ♭VII (the ♭7), secondary dominants (a raised note that resolves up), tritone subs (bass falling by half step).

Transcribing — the fastest ear trainer

- **Choose** something short and slightly too hard: one 4–8 bar phrase.
- **Sing it** until you can sing it without the recording. Only then look for it on the guitar.
- **Find the key and the bass** first, then the melody, then the chords.
- **Slow it down** (software at 50–75%) without changing pitch for fast passages.
- **Analyse** what you transcribed with Part I: chord tones? approach notes? which scale? Then play it in three other keys and use it in your own playing.

A 10-minute daily ear routine

Minutes	Activity
2	Sing a major scale and a minor scale from a random note; check with the guitar.
3	Interval drill: a partner, an app, or the Guitar Cards ear gym — 20 intervals, melodic and harmonic.
3	Chord-quality drill: triads and 7th chords, root position then inversions.
2	Function drill: play I–IV–V–I, then a note or chord; name its degree / numeral.

27 The mastery plan and checklists

How to drill this guide

- **One small thing, every day.** 60 focused minutes beats 4 hours once a week. Short daily sessions consolidate in sleep.
- **Spaced repetition:** revisit each item after 1 day, 3 days, a week, two weeks, a month. Use the Guitar Cards app — it schedules this for you — or a written log.
- **All 12 keys, around the cycle of 4ths** (C F B♭ E♭ A♭ D♭ G♭ B E A D G). One key per day for a shape you know; one key per week for a new one.
- **Slow → fast:** perfect at slow tempo first. Speed is the last thing you add.
- **Say it, then sing it.** Say the interval of every note as you play it; sing the line (or the target note) before you play it.
- **Mix the views:** the same material as a chord, an arpeggio, a scale, a line, in one position and along one string.
- **Make music with it within a week:** every new shape goes into a tune, a progression or an improvisation.

A 60-minute session

Block	Minutes	What	Example
Warm-up / technique	10	scales and arpeggios from Part II, in time	one CAGED position of today's key: scale, then maj7/7/m7 arpeggios, saying intervals
Fretboard & theory	10	note finding, intervals, Part I tables	find every 3rd and 7th of today's ii–V–I on strings 4-3-2
Chords & voice leading	15	triads, drop 2/3, shells through progressions	ii–V–I in today's key with drop 2 on 4-3-2-1, three inversions
Improvisation	15	layers from section 24 over a backing loop	guide tones → arpeggios → approach notes over a blues
Ear	5	section 26 routine	sing the bass line of the tune you are learning
Repertoire / music	5+	play a real tune with everything above	melody, then comp, then a chorus of solo

The mastery ladder (for any item)

Level	You can…
1	play it while reading the diagram
2	play it from memory in C, saying every interval
3	play it in three more keys (G, F, B♭) without hesitation
4	play it in all 12 keys around the cycle of 4ths at 60 bpm
5	play it in time at 120 bpm (8th notes) and start from any note or inversion
6	use it in music: hear it before you play it, and choose it on purpose

Mastery checklist — you have mastered a section when you can…

§	Section	You have mastered it when you can…
1, 2	**Fretboard and intervals**	□ Name any note at any fret in under 2 seconds. □ Find the 3, 5 and ♭7 above any root on any string without counting. □ Sing any interval up and down from any note.
3, 4	**Chord formulas and grips**	□ Spell every chord in the table from any root, aloud, in under 5 seconds. □ Play every open and barre grip, changing between any two in time at 80 bpm. □ Build any chord on the neck from its formula without a stored grip.
5	**Harmonic systems**	□ Name the 7th chord on every degree of major, melodic minor and harmonic minor. □ Name the mode, its formula and its colour note for every degree of all four systems.
6	**Chord–scale**	□ For any chord in a tune, name two scales and play them over it in one position.
7, 8	**Keys and progressions**	□ Write and play the diatonic chords of any key. □ Play ii–V–I, I–vi–ii–V, the three blues forms and rhythm-changes A in all 12 keys. □ Recognise these progressions by ear.
9	**Modal interchange**	□ Name the source and the changed note of every borrowed chord; play all 15 moves in 4 keys.
10	**CAGED**	□ Play a chord, its arpeggio and its scale in all five shapes, in any key, connecting the shapes up and down the neck.
11	**Triads**	□ Play all inversions of all four triad types on all four string sets, in any key, at 60 bpm, naming bass and top note.
12, 13	**Drop 2 and drop 3**	□ Play every inversion of maj7, 7, m7, m7♭5 on each string set in any key. □ Comp a ii–V–I in any key with the nearest inversion each time.
14, 15	**Shells, grips, harmonised scale**	□ Comp a jazz blues and rhythm changes with shells only; then add 9ths, 13ths and altered grips. □ Play the diatonic 7th chords of any key up the neck on two string sets, naming each numeral.
16, 17, 18, 19, 20, 21	**Scales and arpeggios**	□ Play the major scale in all CAGED and 3NPS shapes in any key; sequence it in 3rds and 4ths. □ Play all seven modes from one root and hear each colour note. □ Play all five pentatonic boxes connected end to end. □ Play every arpeggio in all five shapes from any chord tone. □ Locate melodic minor, harmonic minor, diminished and whole-tone shapes around any root.
22, 23, 24	**Voice leading, reharmonisation, improvisation**	□ Play a guide-tone line through a standard. □ Arrange a simple melody as chord melody, then reharmonise two bars of it. □ Improvise over ii–V–I and blues landing chord tones on the changes, with enclosures and a developed motif.

§	Section	You have mastered it when you can…
25, 26	**Rhythm and ear**	□ Play with the click on 2 and 4 at 60 and 200 bpm; keep time through a 4-bar gap. □ Transcribe and play an 8-bar solo phrase; identify a progression's numerals by ear.